People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research
BADJI Mokhtar University Annaba
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Muscular system

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Introduction

1. Definition of Muscle

The muscle, the driving force of the locomotor system, is an organ that contracts (shortens) under the influence of a stimulus (nerve, electrical, thermal, painful excitation, etc.). The study of muscles is Myology.

2. Roles of the muscular system

- Production of the movement
- Heat production (thermogenesis)
- Maintaining posture
- Joint stabilization
- Protection of the viscera
- Visceral functions : Digestion, Defecation, Urination, Blood propulsion, Vasoconstriction, Vasodilation, Childbirth, etc.

3. Properties of muscles

- Excitability and Contractility: it is the ability to contract following an excitation;
- Elasticity: it is the ability to stretch (without being torn);
- Tone: This is the ability to maintain a slight, permanent muscle contraction, despite its relaxation. It allows us to maintain balance at rest (posture), and disappears upon lying down, in a coma, and in death (atonic subject).

4. Muscle classification

There are three types of muscles…

	Skeletal Muscle	Cardiac Muscle	Visceral muscle
Other names	Striped, Red	Mixed	Smooth
Cytology	Striated cells	Striated cells	Non-striated cells
Location	Attached to the skeleton	Heart walls	hollow viscera Ship walls Skin
Muscle contraction	Voluntary Fast	Involuntary Automatic Rhythmic Fast	Involuntary Slow Continue
Fatigability	Fast	Never	Never
Nervous control by :	Central nervous system (CNS)	Autonomic nervous system (ANS)	Autonomic nervous system (ANS)

Skeletal muscle

1. Number

There are : 600 constant skeletal muscles.

This number varies due to the existence of inconstant muscles, which are accessory muscles (Ex : plantaris muscle).

2. Situation

Two muscle groups :

- Deep muscles (sub-aponeurotic). Ex: muscles attached to the digestive and respiratory systems, skeletal muscles, muscles attached to the sensory organs;
- Superficial muscles (supra-aponeurotic, subcutaneous), Ex: facial muscles.

3. Division

A simple muscle is a monogastric muscle, composed of three parts:

- A middle section : called the muscular belly (or body), it is large, red, fleshy and contractile;
- Two ends : called muscle tendons, they are narrow, whitish, elastic and resistant.

Besides this division, muscle can present four varieties:

- Digastric muscle: it is a muscle which has two bellies and an intermediate tendon. Ex : digastric muscle;
- Multigastric muscle: it is a muscle that has several bellies and several intermediate tendons. Ex: rectus abdominis muscle;
- Bifid muscle: it is a muscle that has two bellies that are arranged in parallel. Ex : biceps brachii muscle;
- Multifidus muscle: It is a muscle that has several bellies arranged in parallel. Ex : quadriceps femoris muscle.

Noticed: In the case of a bifid or multifid muscle, the parallel tendons are called heads.

4. Insertion

The muscle attaches via a double muscular insertion:

- Original Insertion, called proximal or fixed;
- Inserting the termination, called distal or mobile.

Noticed: Sometimes, the original insertion can become mobile.

Muscle insertion can occur at four sites:

- Bone insertion. Ex : iliopsoas muscle ;
- Cartilage insertion. Ex : cricothyroid muscle ;
- Aponeurotic insertion. Ex: abductor hallucis muscle;
- Skin insertion. Ex : palmaris brevis muscle.

5. Shape

- Long muscle: its length predominates over its width and thickness. Ex: sartorius muscle;
- Flat muscle: its length and width predominate over its thickness. Ex : latissimus dorsi muscle ;
- Short muscle: the three dimensions are reduced. Ex: muscles of the hypothenar compartment;
- Annular muscle: is located around the openings. Ex: orbicularis oculi muscle of the eyelids, orbicularis oris muscle of the lips;
- Geometric muscle : Ex : trapezius muscle, pronator quadratus muscle, depressor anguli oris muscle.

6. Summary : Nomenclature

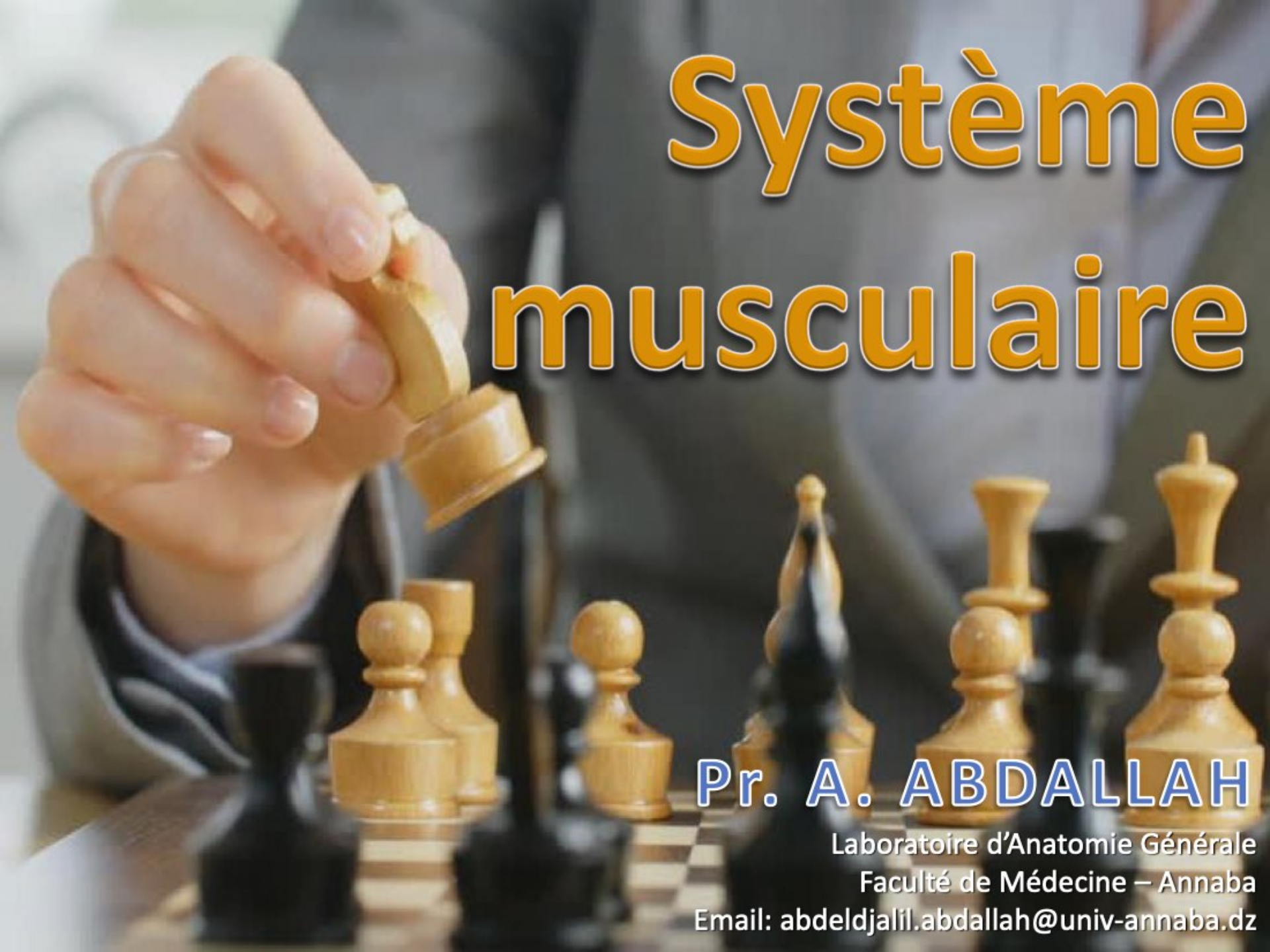
Muscle nomenclature is based on morphological and functional criteria.

- Size criterion. Ex : gluteus maximus muscle, pectoralis minor muscle, fibularis longus muscle ;
- Action criterion. Ex: adductor magnus muscle, supinator muscle, superficial flexor muscle of the fingers;
- Situational criterion. Ex : dorsal interosseous muscles, gluteal muscles, brachialis muscle;
- Form criterion. Ex : piriformis muscle, quadratus plantaris muscle ;
- Muscle fiber direction criterion. Ex : internal oblique muscle, rectus abdominis muscle ;
- Insertion criteria. Ex : brachioradialis muscle, coracobrachialis muscle ;
- Number of heads criterion. Ex: biceps femoris muscle, triceps brachii muscle.

7. Appendices

These are anatomical structures that are useful to the muscle, facilitating its action. There are four:

- Serous (synovial) bursae: these are membranous sacs filled with synovial fluid, located near the joints and facilitate the gliding of muscles;
- Tendinous serous sheaths: these are serous bursae surrounding the tendons;
- Fibrous sheaths: these are fibrous grooves, which delimit with the bone an osteofibrous canal, where the tendons slide, enveloped by serous sheaths;
- Muscular fascia (covering aponeuroses): These are fibrous membranes, that encase and separate the muscles.



Système musculaire

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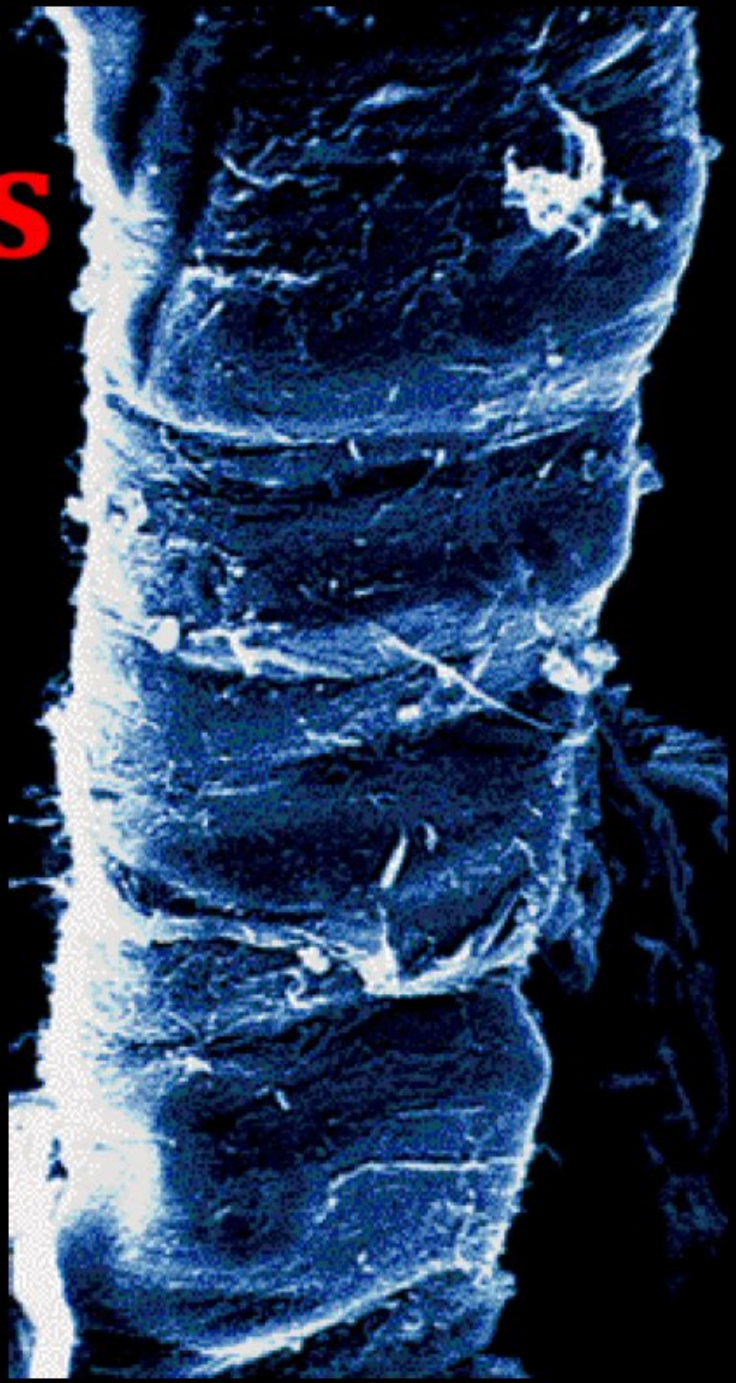
Un système musculaire…

… fort, précis et intelligent



Plan du cours

- Introduction
- Muscle squelettique
- Myologie clinique



Introduction



Définition du muscle

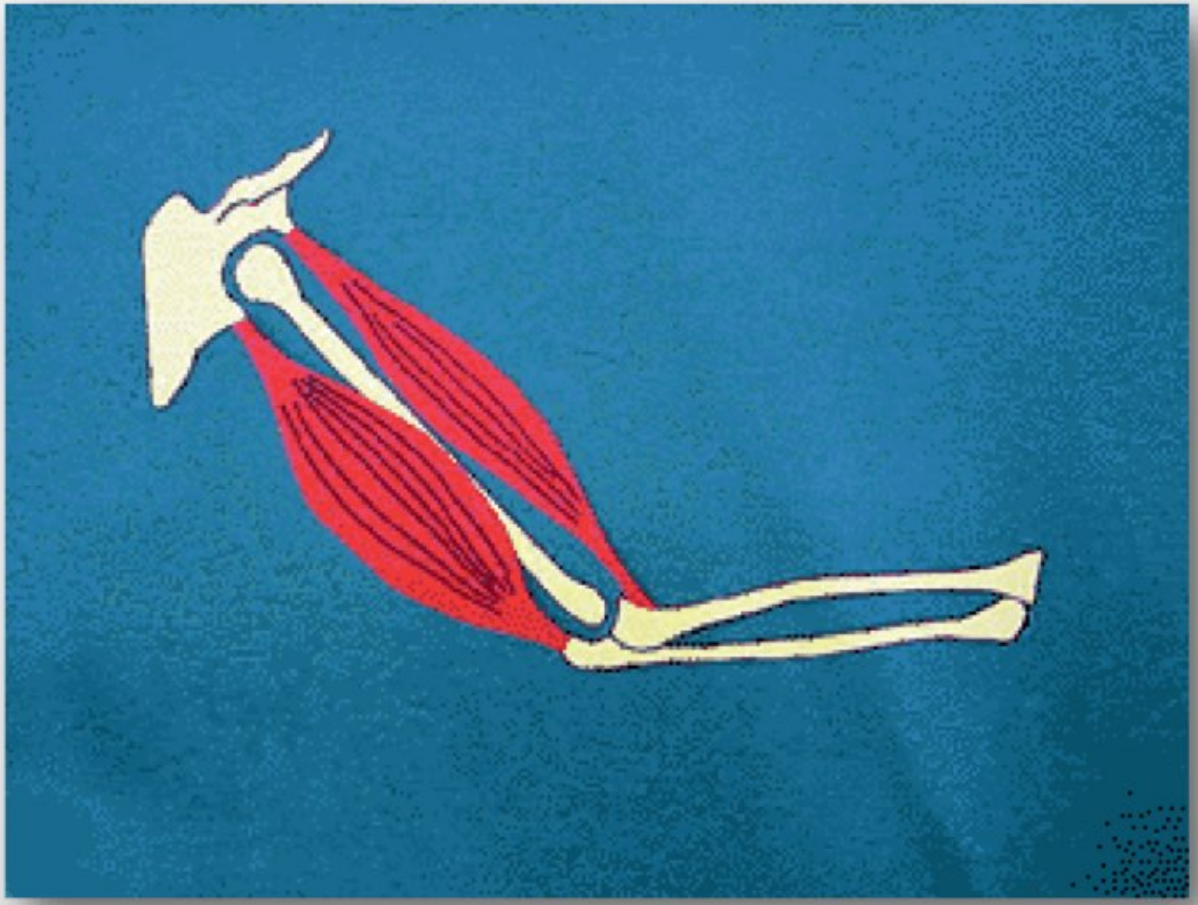
Rôles du système musculaire

Propriétés des muscles

Classification des muscles

Le muscle est:

l'élément moteur de **l'appareil locomoteur**



= organe qui

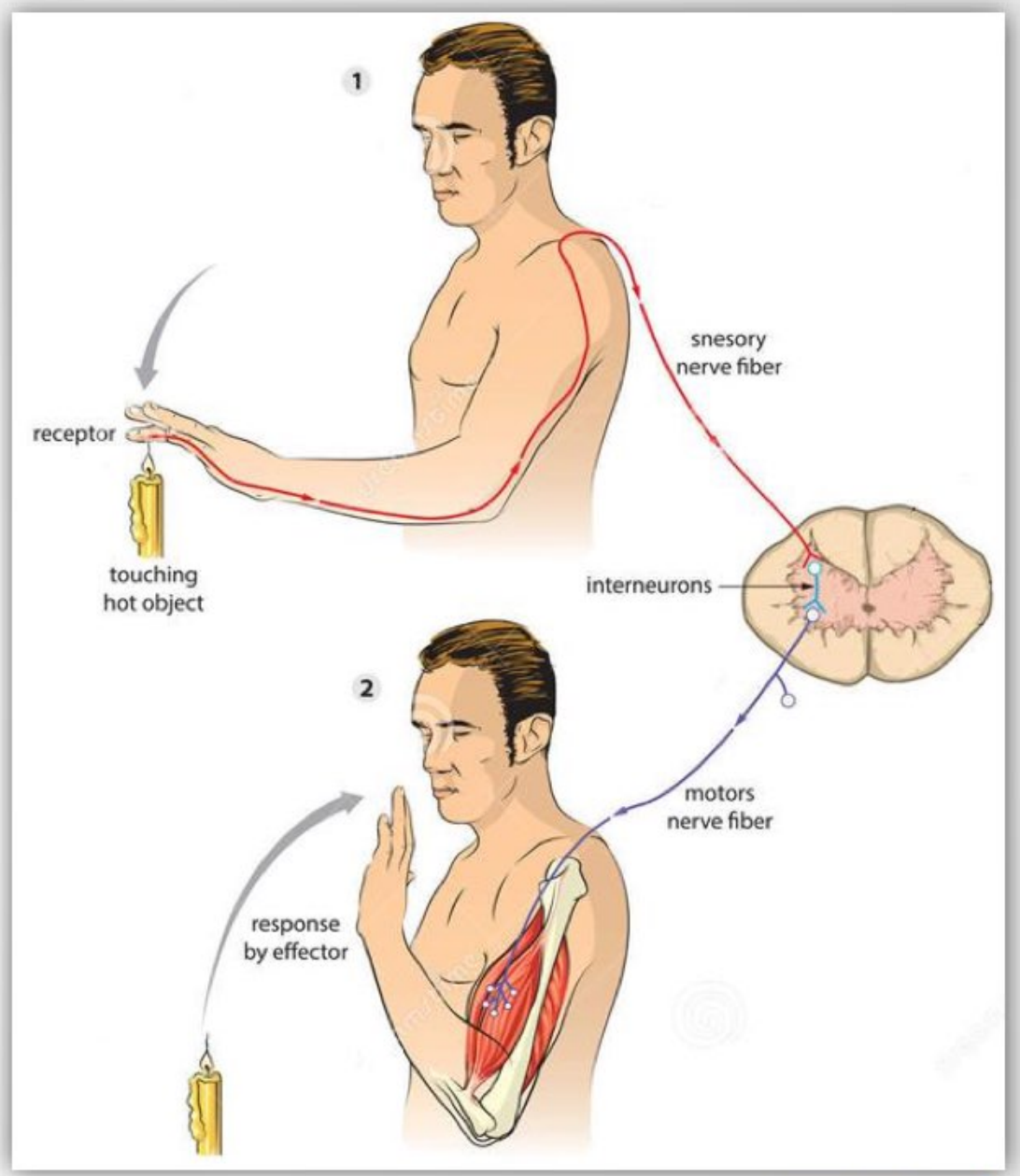
se contracte

(se raccourcit)

sous l'influence

d'un **stimulus**

(excitation nerveuse, électrique, thermique, douloureuse …)



- **L’étude des muscles** est **la myologie**

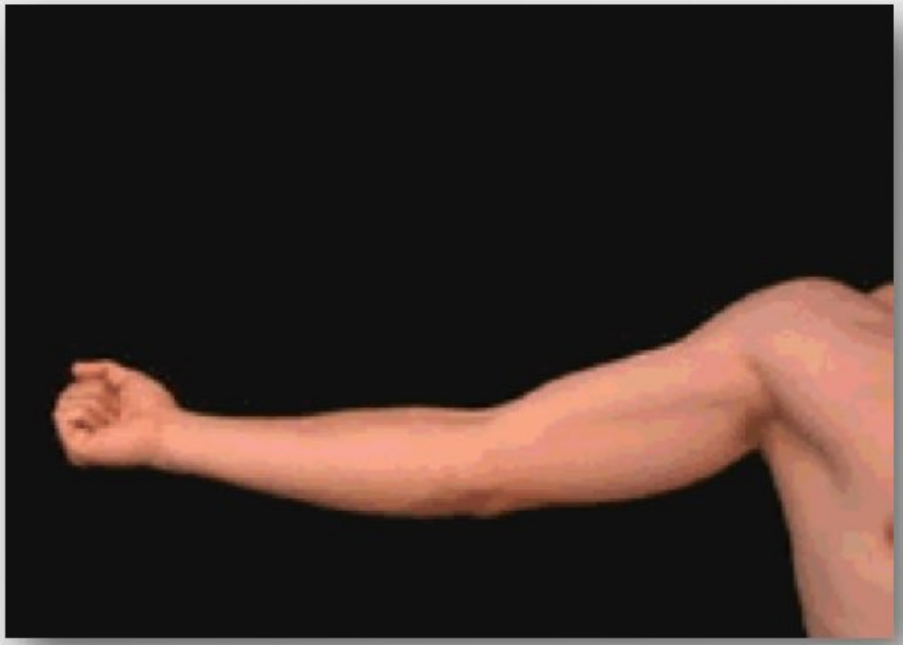


Définition du muscle

Rôles du système musculaire

Propriétés des muscles

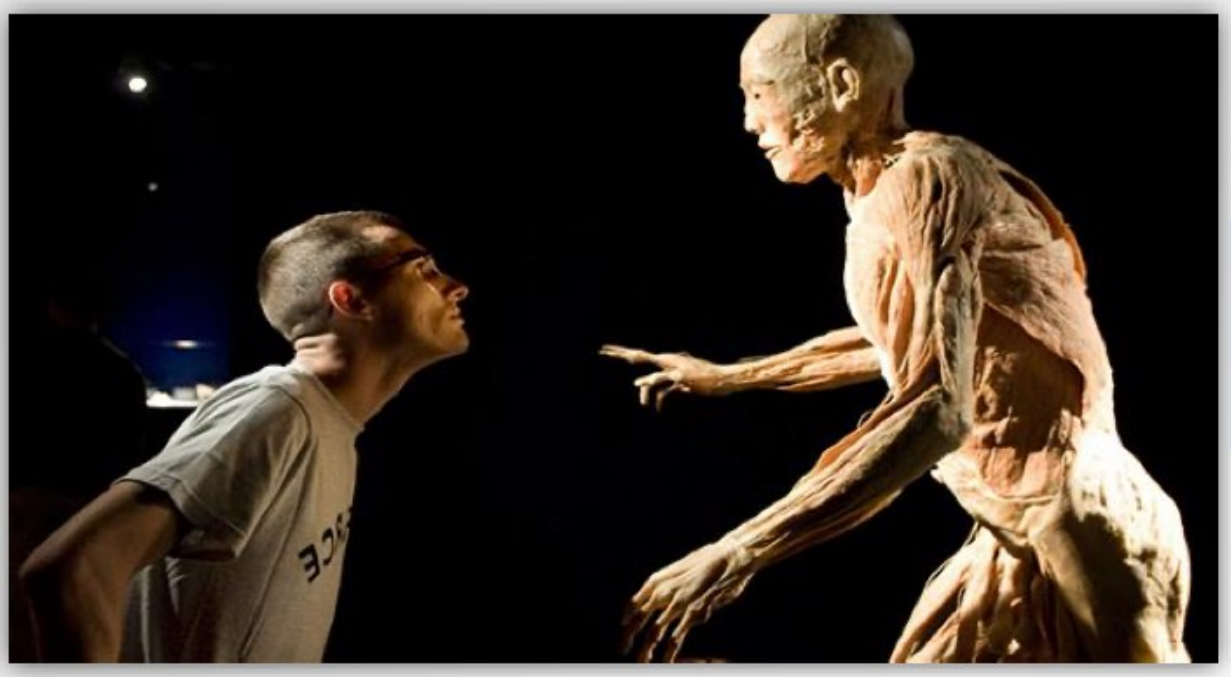
Classification des muscles



(1) Production du **mouvement**



(2) Production de **la chaleur** (**thermogenèse**)



(3)

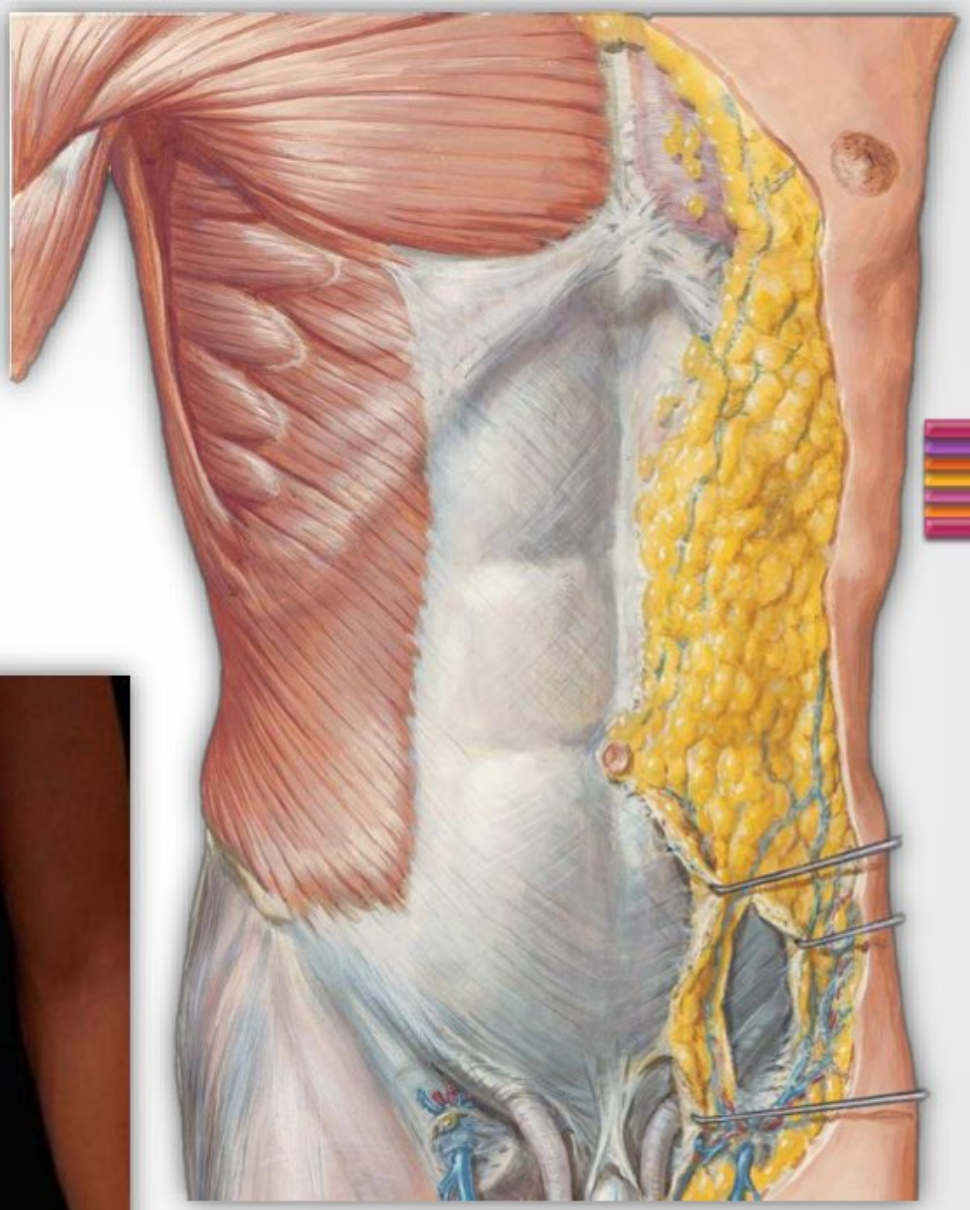
Maintien
de **la posture**

(4)

Stabilisation
des **articulations**



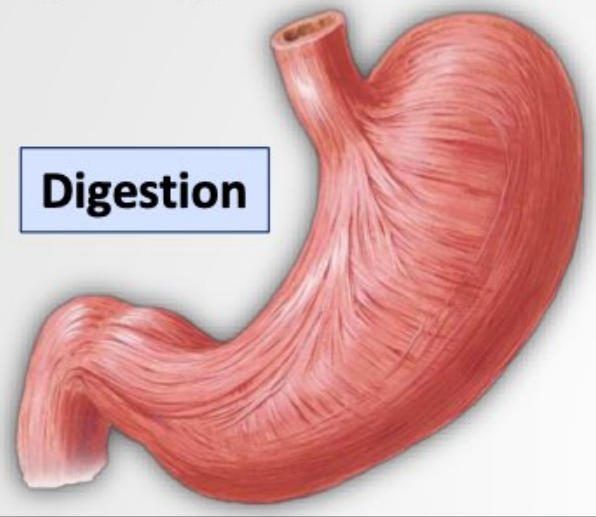
(5) Protection des **viscères**

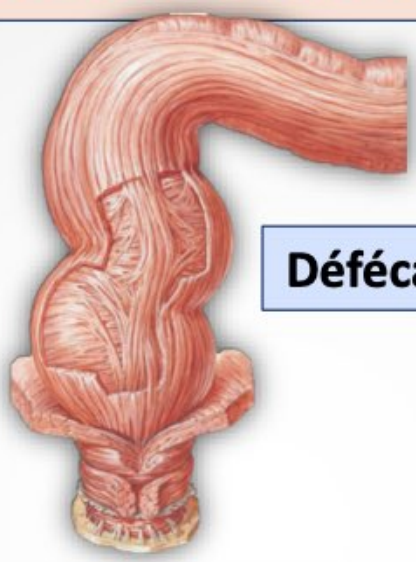




(6) Fonctions viscérales:

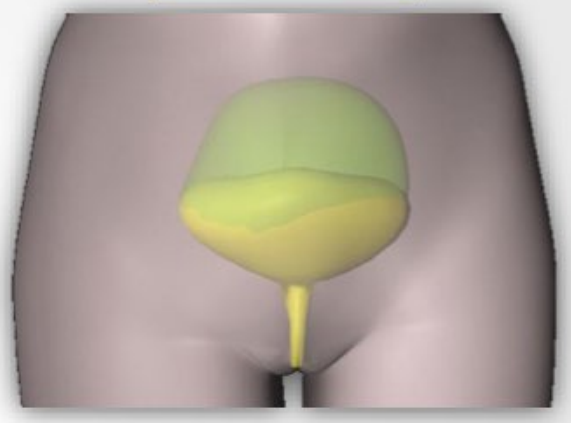
Digestion



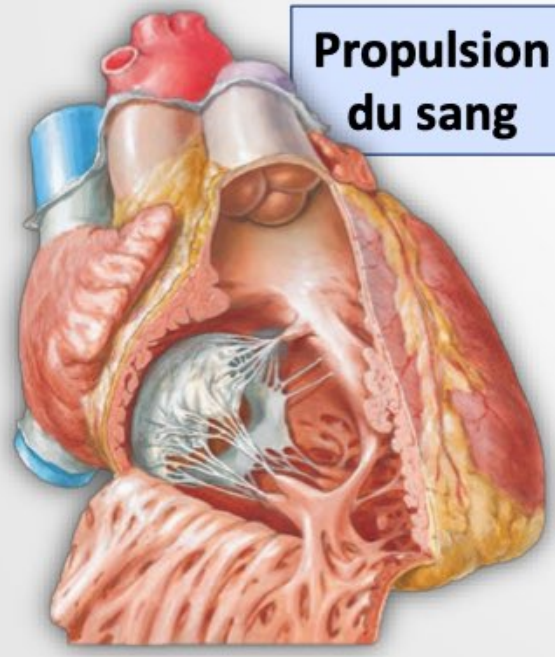


Défécation

Miction

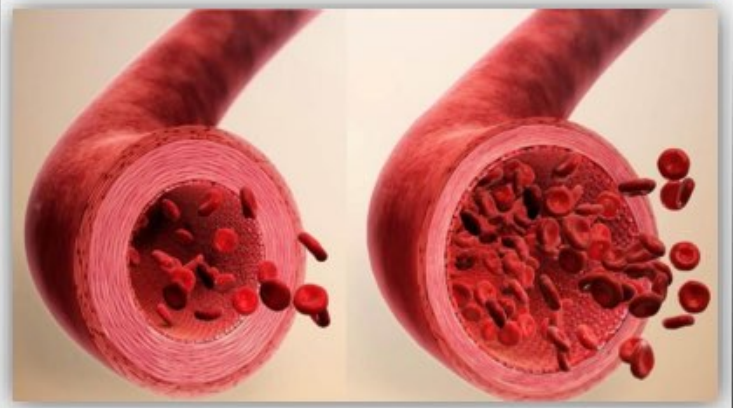


Propulsion du sang

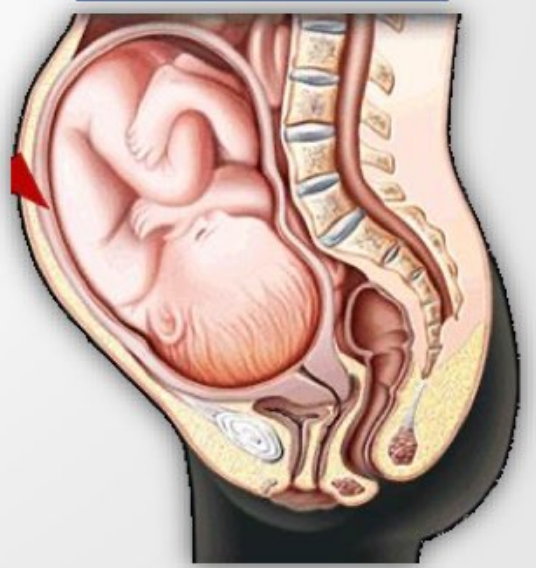


Vasoconstriction

Vasodilatation



Accouchement



Définition du muscle

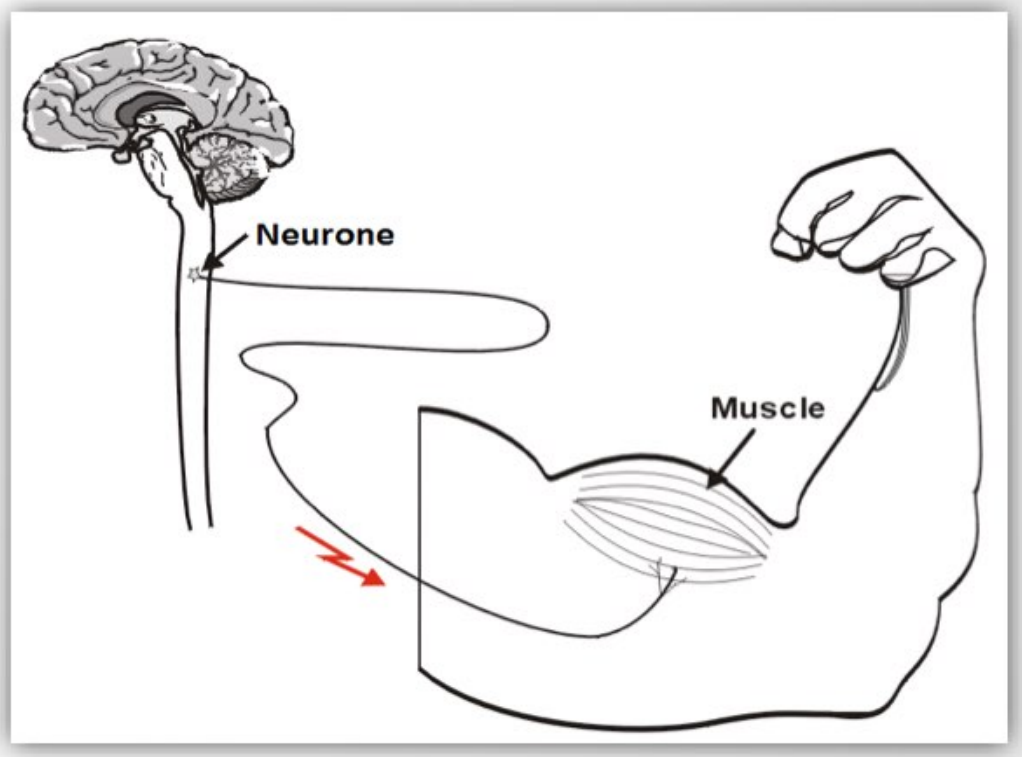
Rôles du système musculaire

Propriétés des muscles

Classification des muscles

(1, 2) Excitabilité et Contractilité:

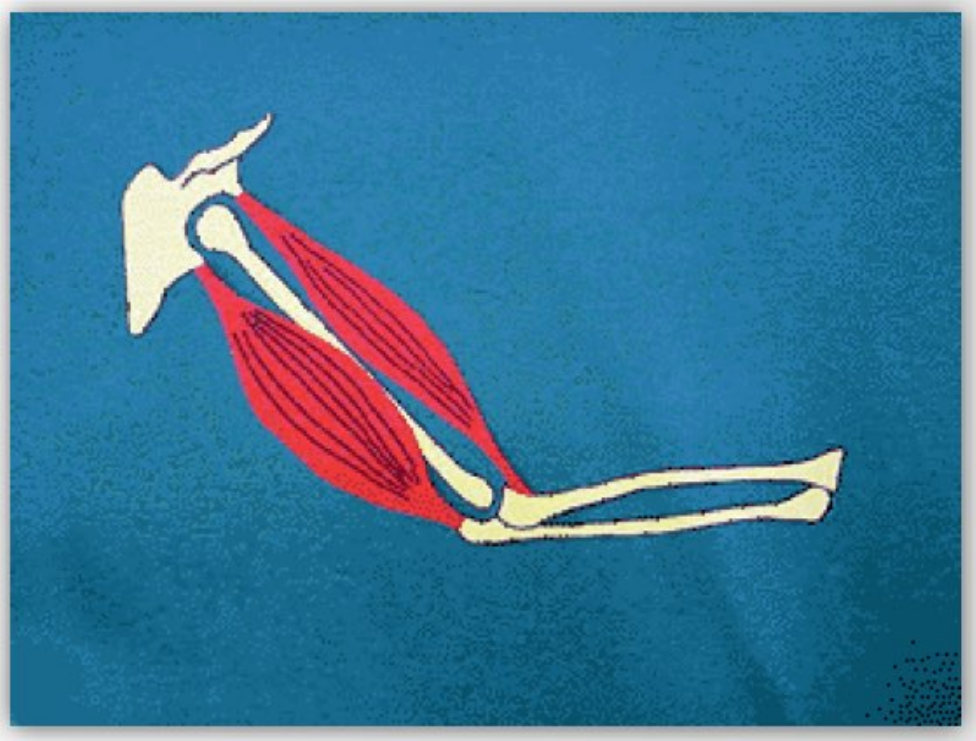
= capacité de **se contracter** suite à une **excitation**



(3) Élasticité:

= capacité de **s'étirer**

(sans être déchiré)



(4) Tonicité :

= capacité de **conserver** une **légère contraction permanente du muscle** (= **tonus**) malgré son relâchement



Disparait lors du:
coucher, coma, mort
(sujet atonique)

Permet de **se maintenir en équilibre au repos** (**posture**)





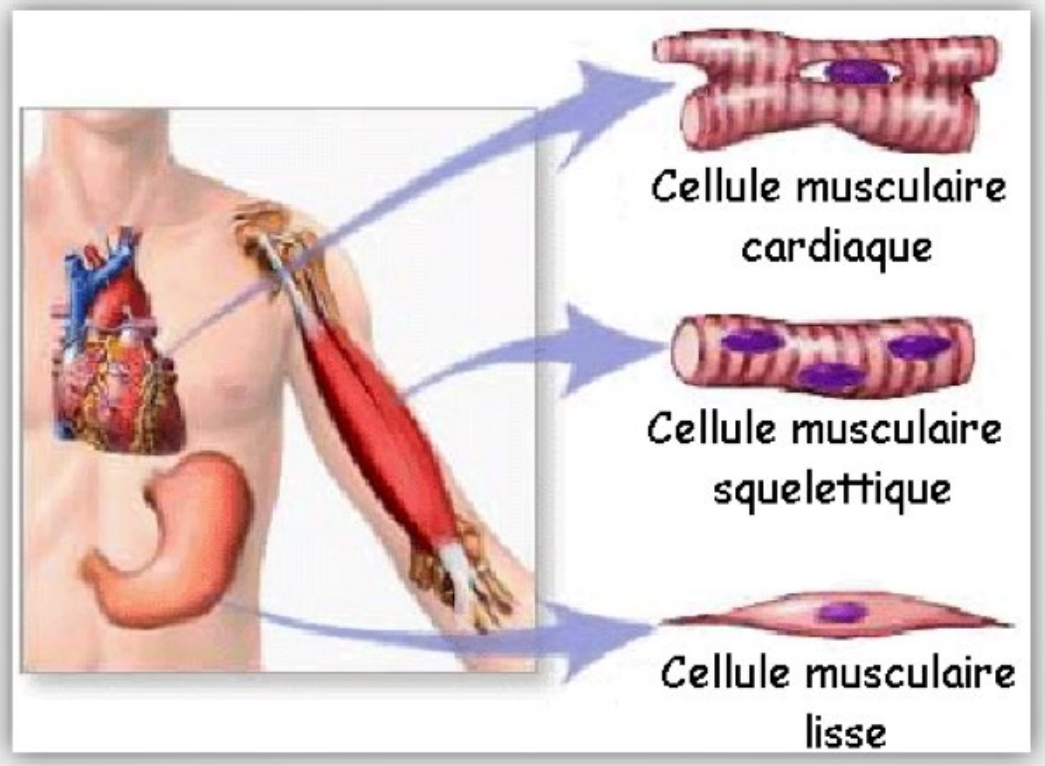
Définition du muscle

Rôles du système musculaire

Propriétés des muscles

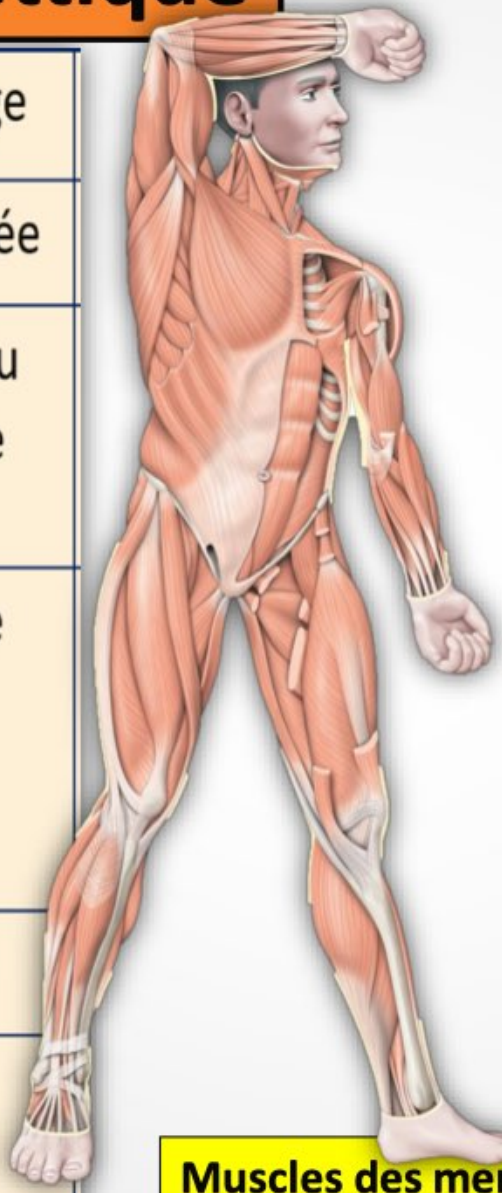
Classification des muscles

3 types musculaires…

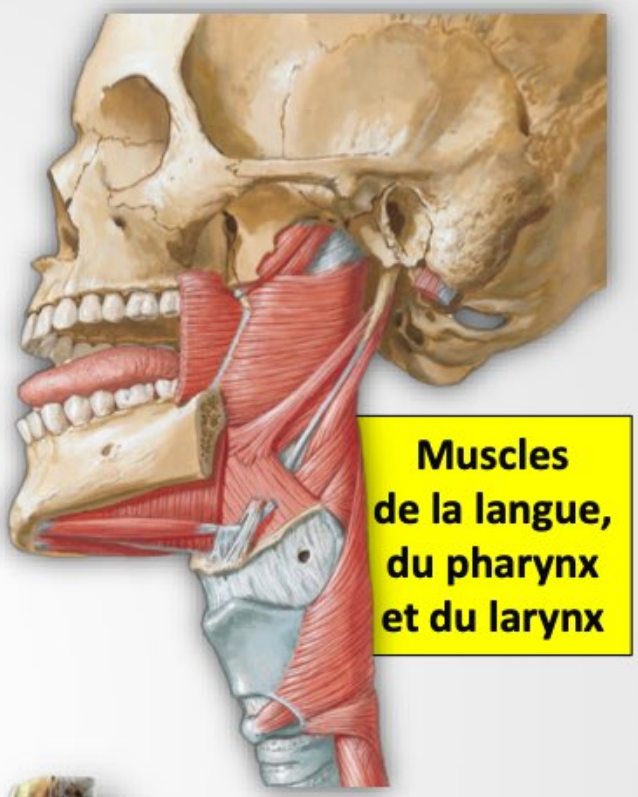


Muscle squelettique

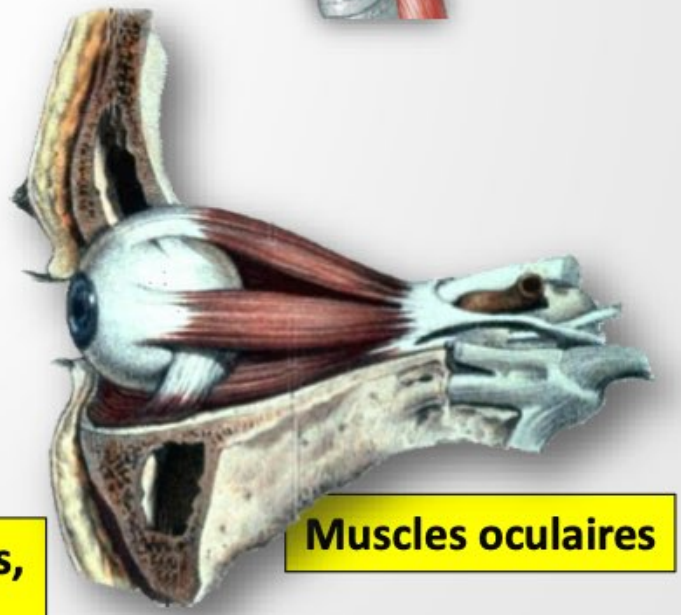
Autres noms	Strié, rouge
Cytologie	Cellule striée
Localisation	Attaché au squelette
Contraction musculaire	- Volontaire - rapide
Fatigabilité	Rapide
Contrôle nerveux par:	SNC



Muscles des membres, tronc, tête et cou



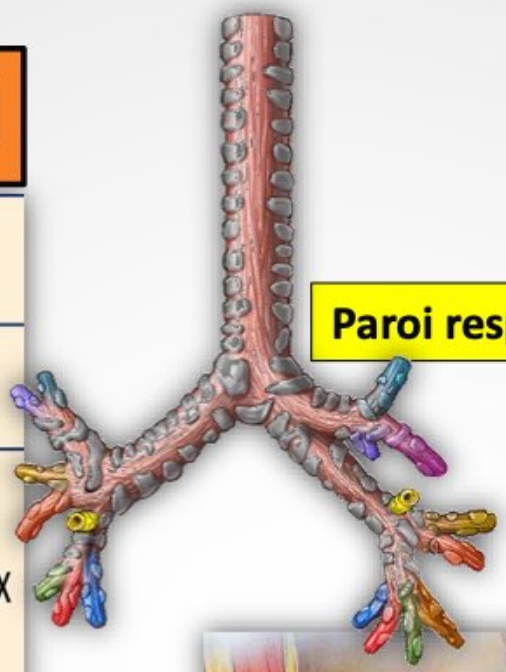
Muscles de la langue, du pharynx et du larynx



Muscles oculaires

Muscle viscéral

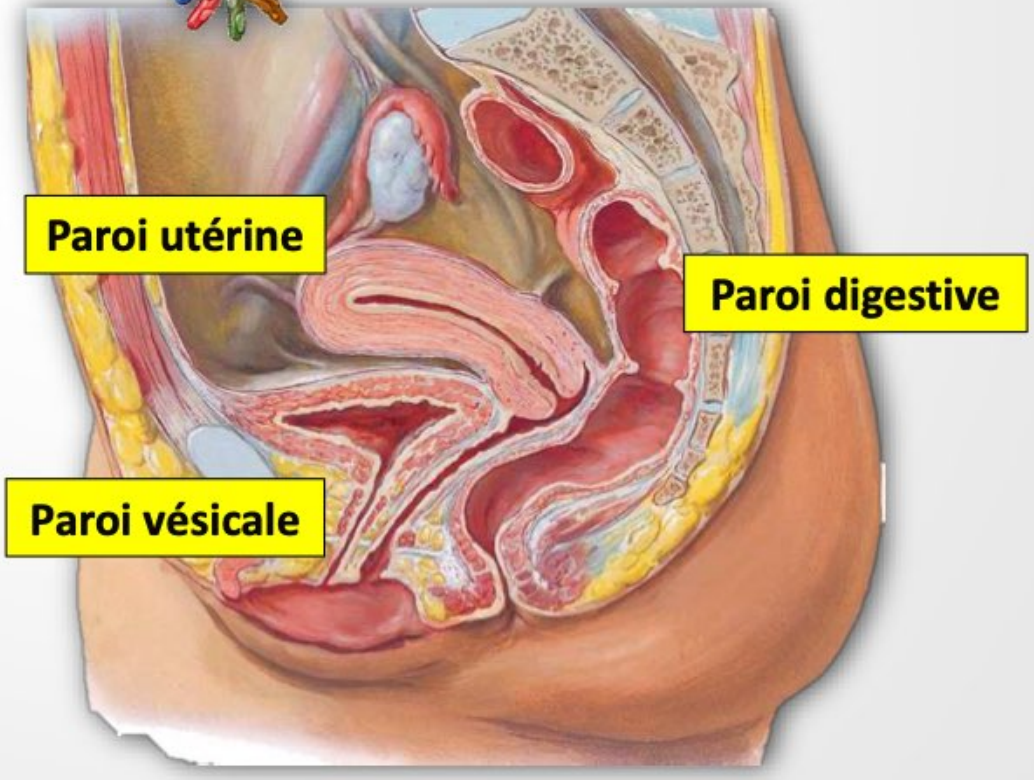
Autres noms	Lisse
Cytologie	Cellule non striée
Localisation	- Viscères creux - Parois des vaisseaux - Peau
Contraction musculaire	- Involontaire - Lente - Continue
Fatigabilité	Jamais
Contrôle nerveux par:	SNV



Paroi respiratoire



Paroi vasculaire



Paroi utérine

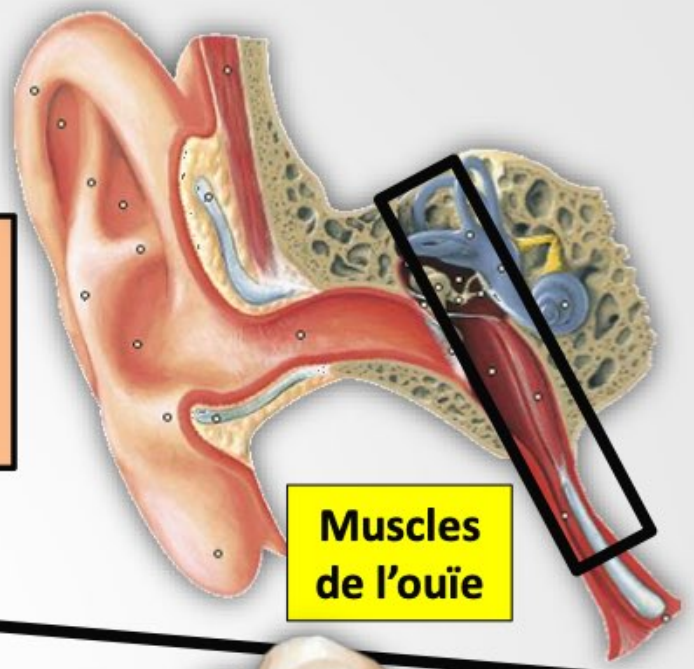
Paroi digestive

Paroi vésicale

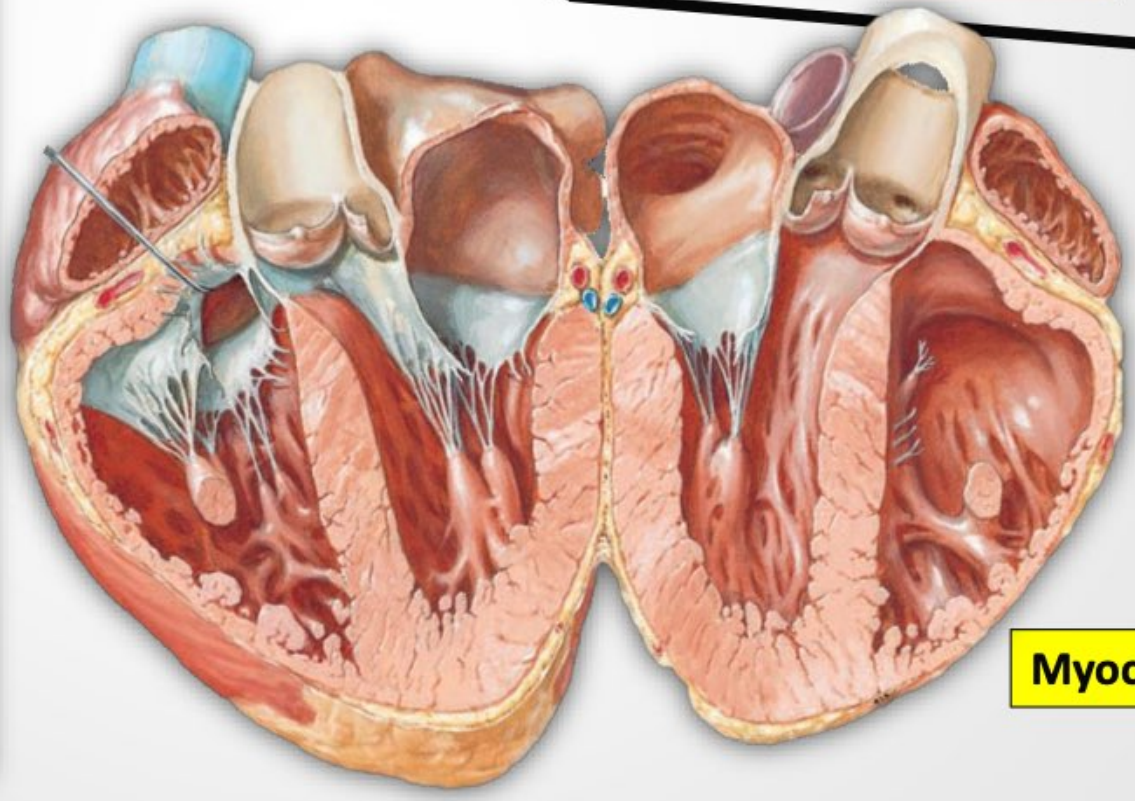
Muscle cardiaque

Autres noms	Mixte
Cytologie	Cellule striée
Localisation	Parois du cœur
Contraction musculaire	- Involontaire - Automatique - Rythmique - Rapide
Fatigabilité	Jamais
Contrôle nerveux par:	SNV

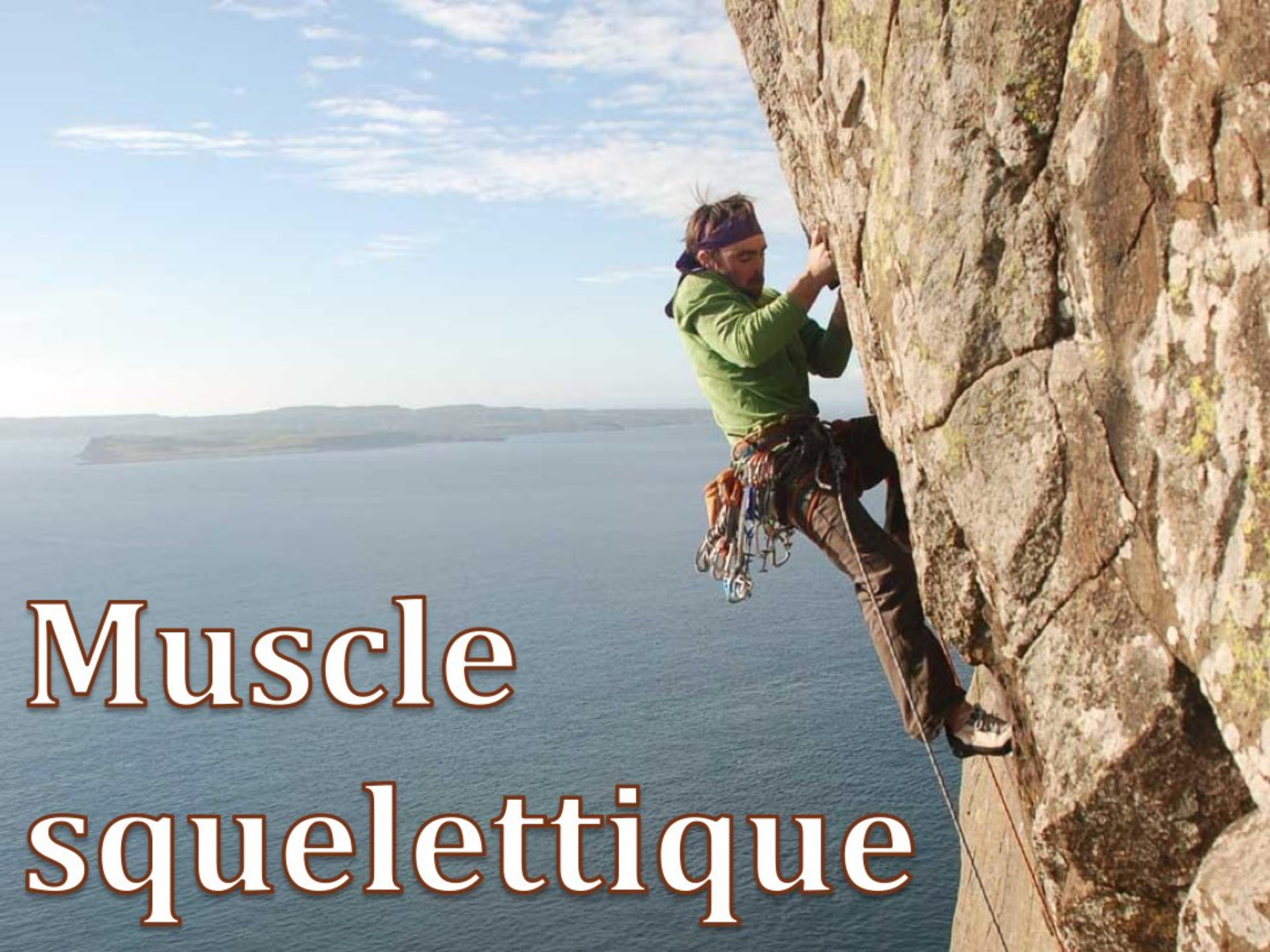
Un autre **muscle mixte**

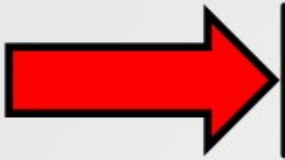


Muscles de l'ouïe



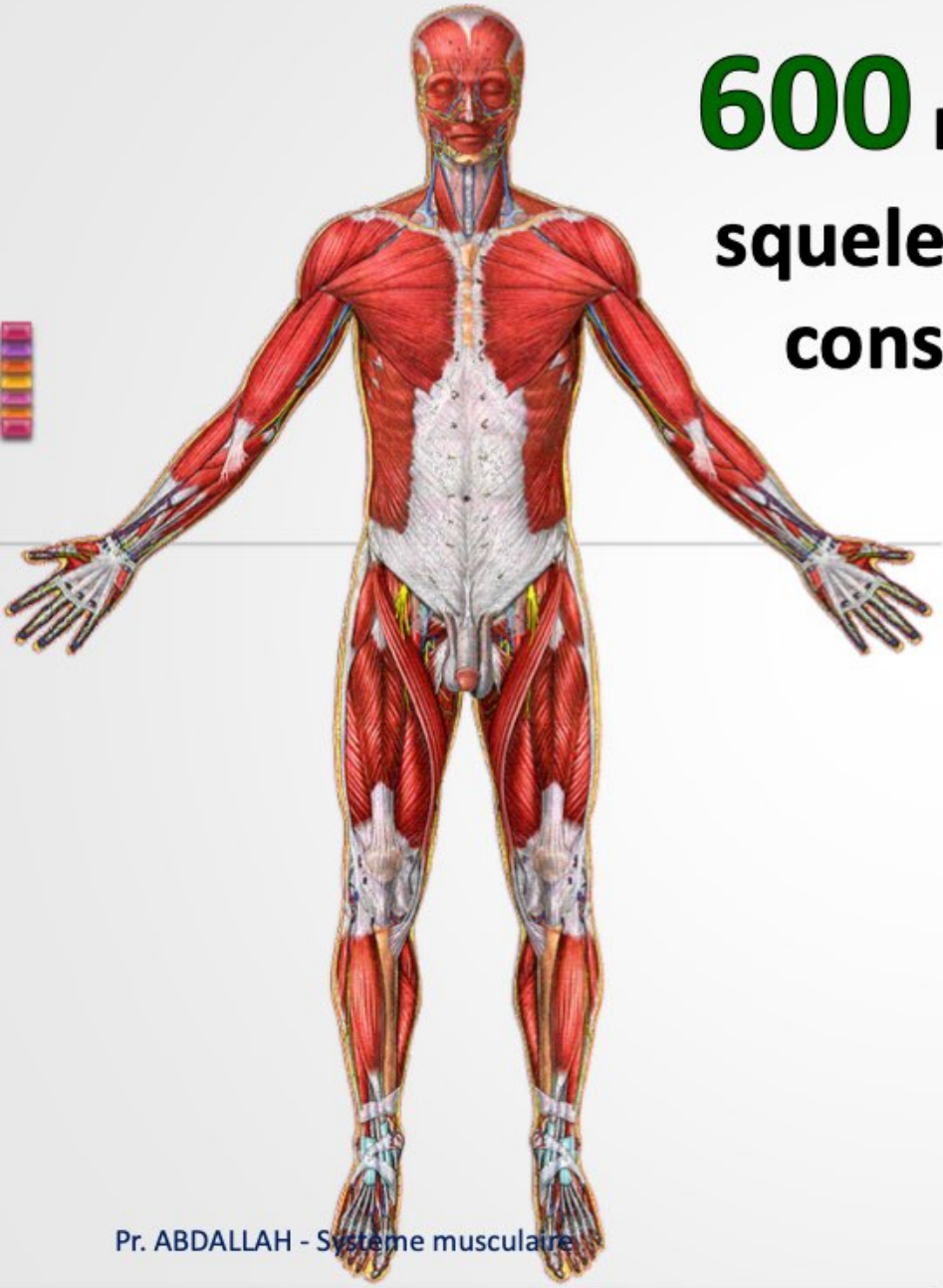
Myocarde

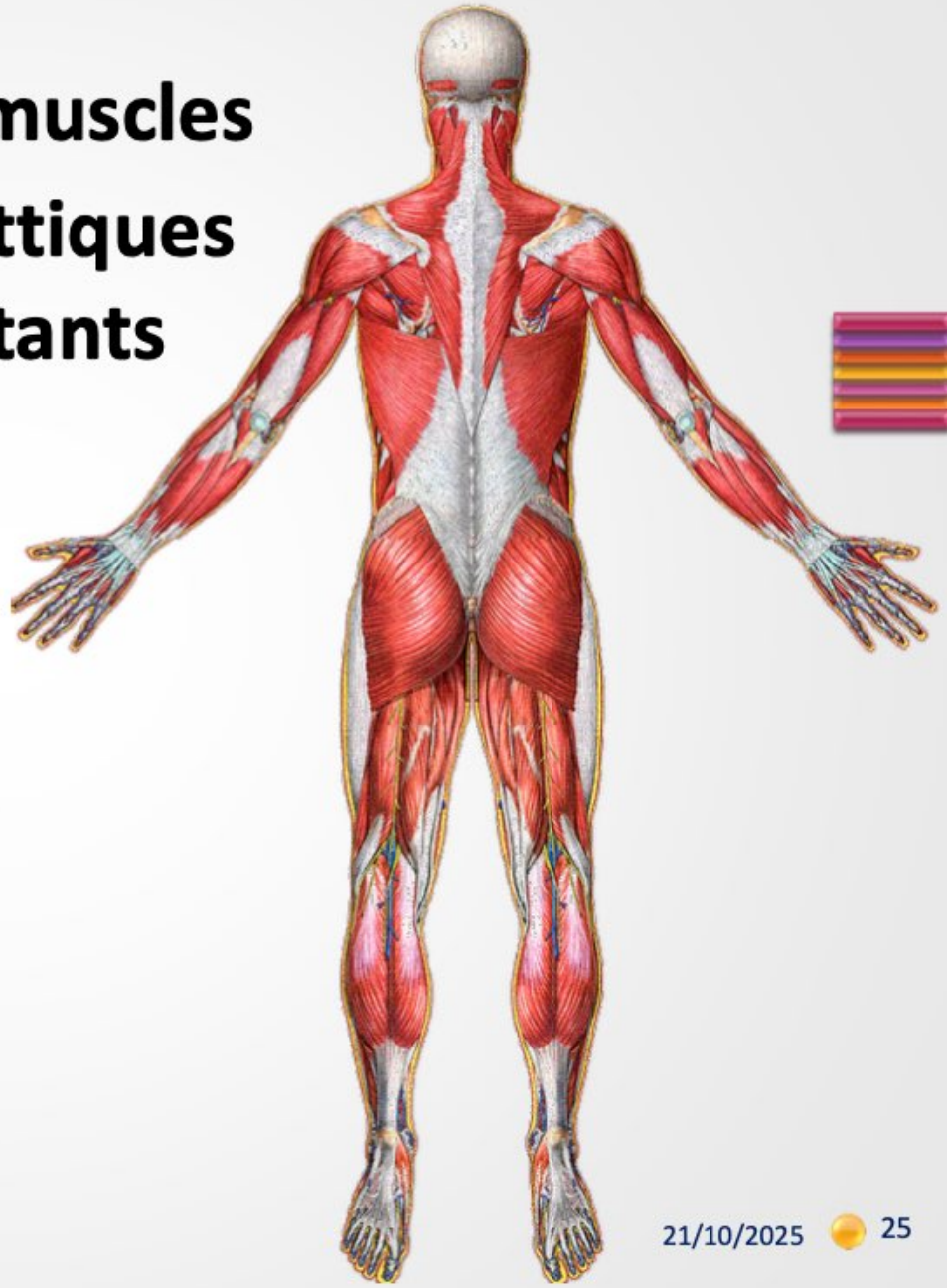

Muscle
squelettique



- Nombre
- Situation
- Division
- Insertion
- Forme
- Synthèse: Nomenclature
- Annexes

600 muscles squelettiques constants





Remarque:

- Ce nombre est **variable** par existence de **muscles inconstants**

➔ **Muscles accessoires**



Nombre

Situation

Division

Insertion

Forme

Synthèse: Nomenclature

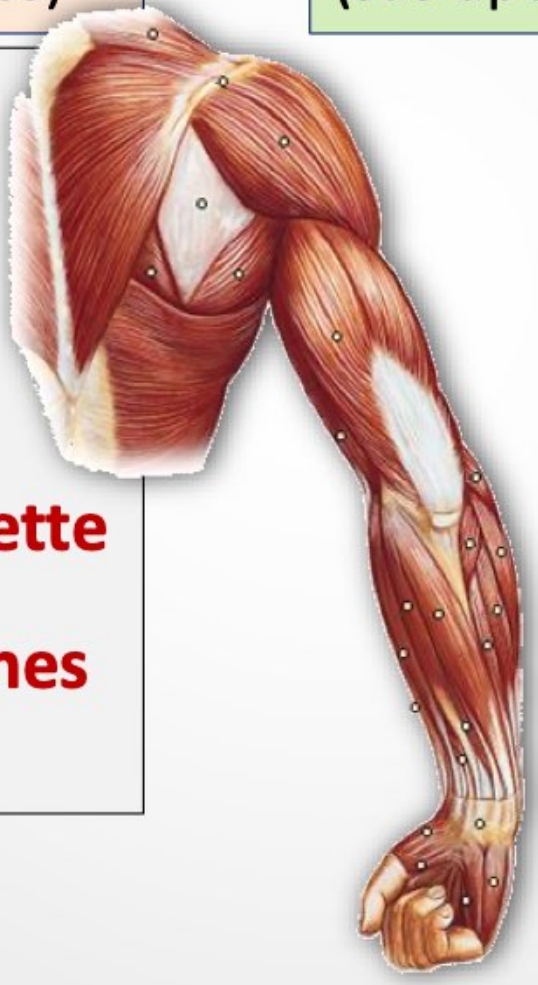
Annexes

2 groupes musculaires…

(1) Muscles profonds
(sous-aponévrotiques)

Ex:

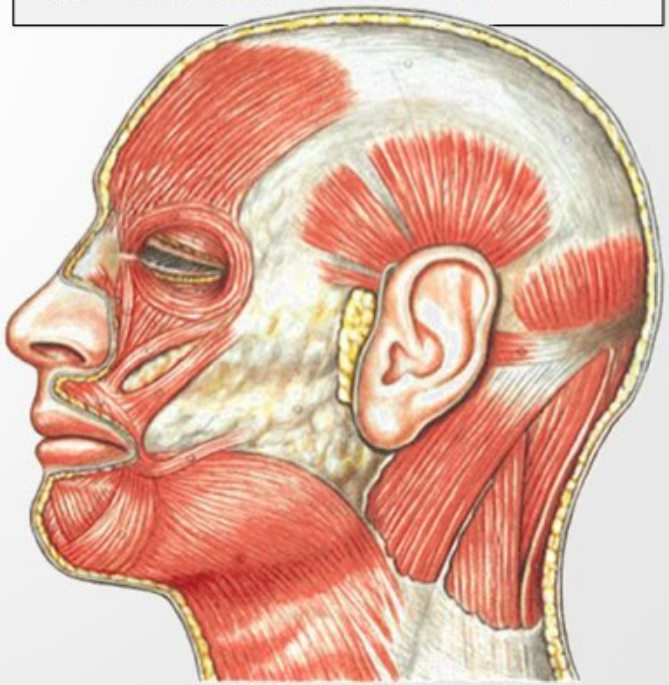
- Muscles des **appareils digestif** et **respiratoire**
- Muscles du **squelette**
- Muscles des **organes de sens**



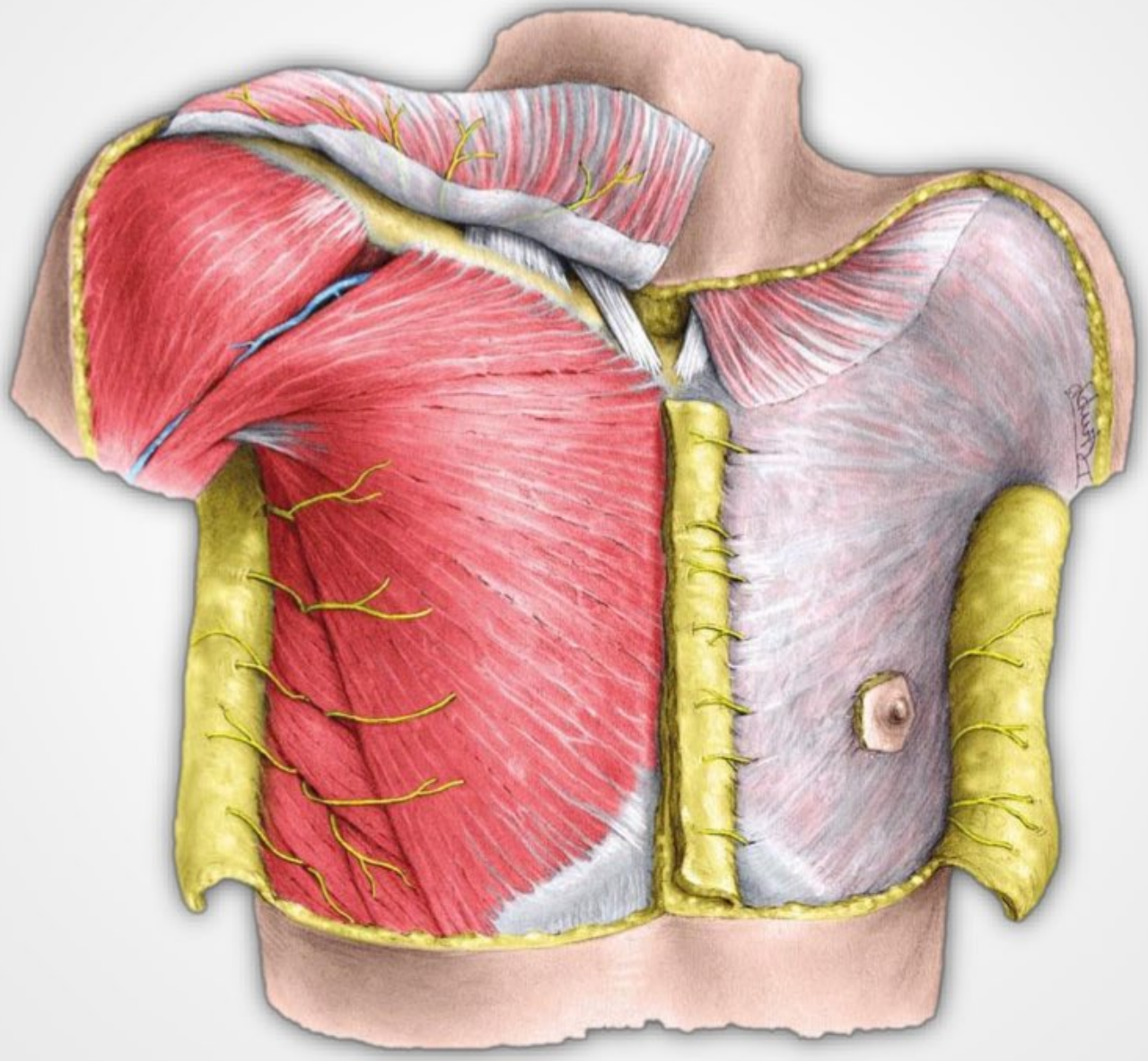
(2) Muscles superficiels
(sus-aponévrotiques, sous-cutanés)

Ex:

muscles peauciers de la tête et du cou



Aponévrose et Muscles sous-aponévrotiques



Nombre

Situation

Division

Insertion

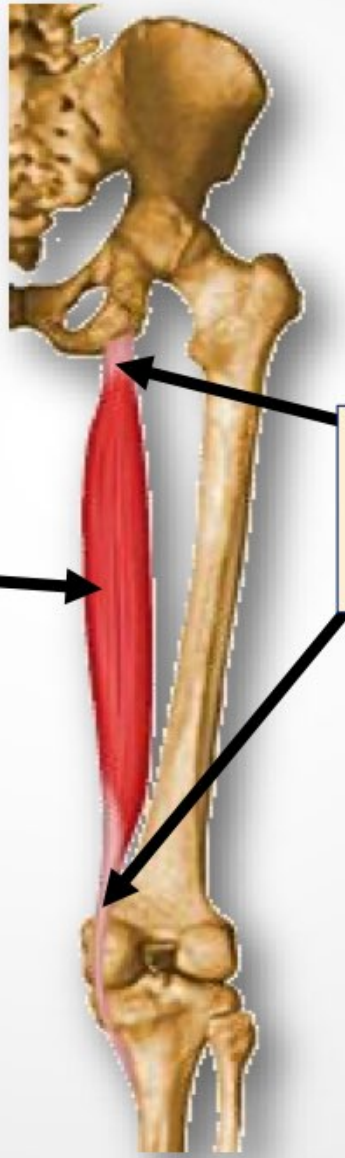
Forme

Synthèse: Nomenclature

Annexes

Un **muscle simple** = **monogastrique**

Formé de **3** **parties…**



1 Partie médiane:

Ventre (**corps**) **musculaire**

Volumineux
Rouge
Charnu
Contractile

2 extrémités:

Tendons musculaires

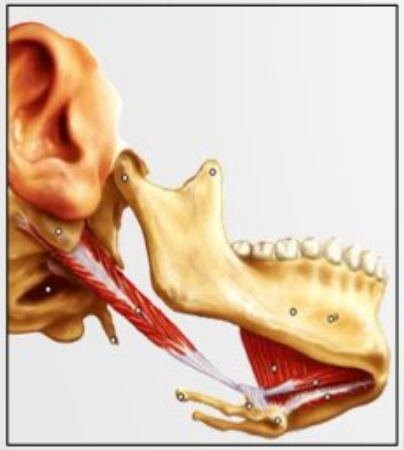
Étroits
Blanchâtres
Élastiques
Résistants

Cette **division** présente:

4 variétés

(1) Muscle digastrique

2 ventres
+ 1 tendon intermédiaire



Digastrique

(2) Muscle multigastrique

Plusieurs ventres + plusieurs tendons intermédiaires



Droit de l'abdomen

(3) Muscle bifide

2 ventres
se disposent en parallèle



Biceps brachial

(4) Muscle multifide

Plusieurs ventres
se disposent en parallèle

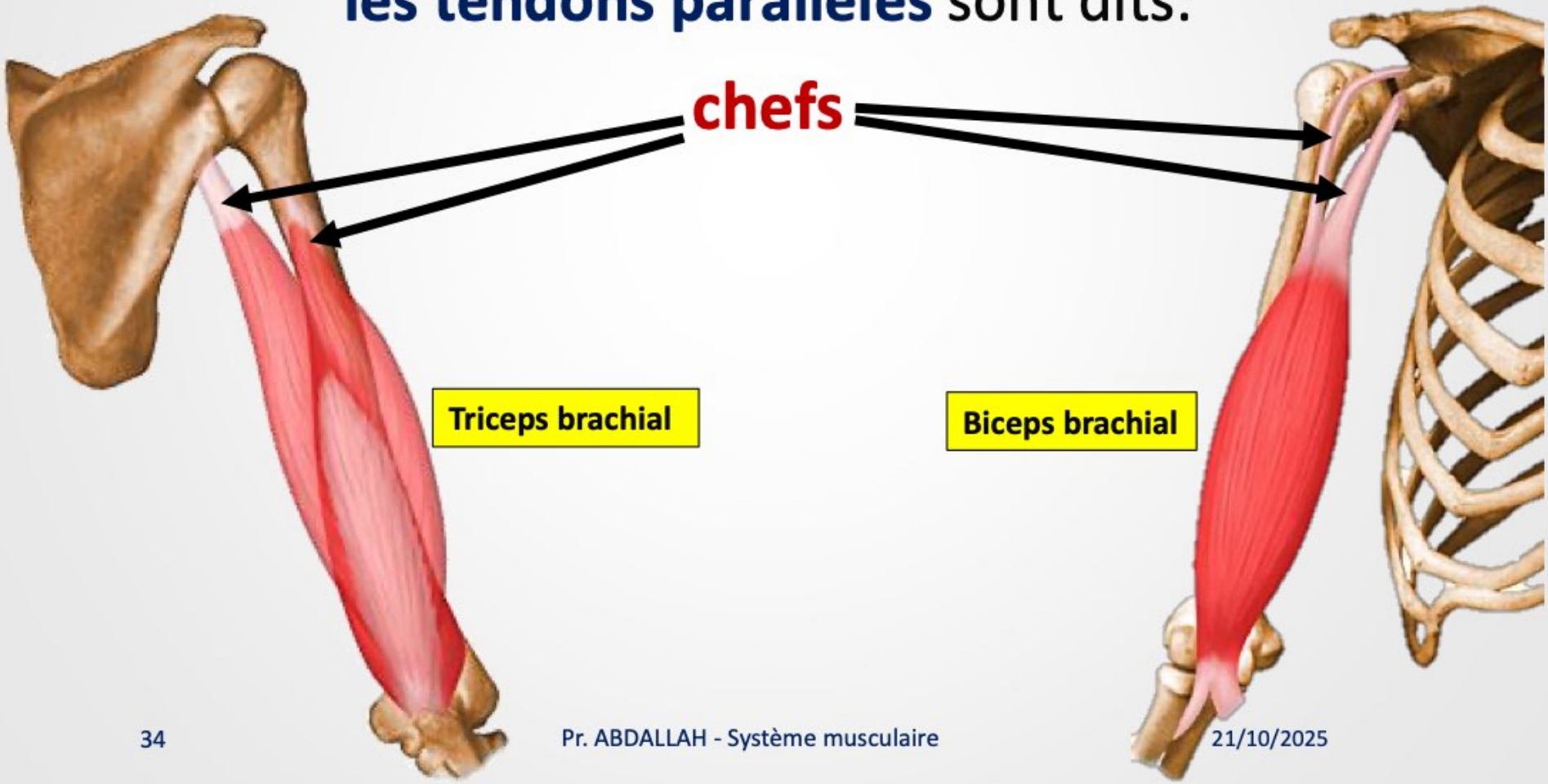


Quadriceps fémoral

Remarque

Dans le cas d'un **muscle bifide** ou **multifide**, **les tendons parallèles** sont dits:

chefs



Nombre

Situation

Division



Insertion

Forme

Synthèse: Nomenclature

Annexes

On distingue: une **double** insertion musculaire…



Muscle brachioradial

Insertion d'origine

= proximale = fixe

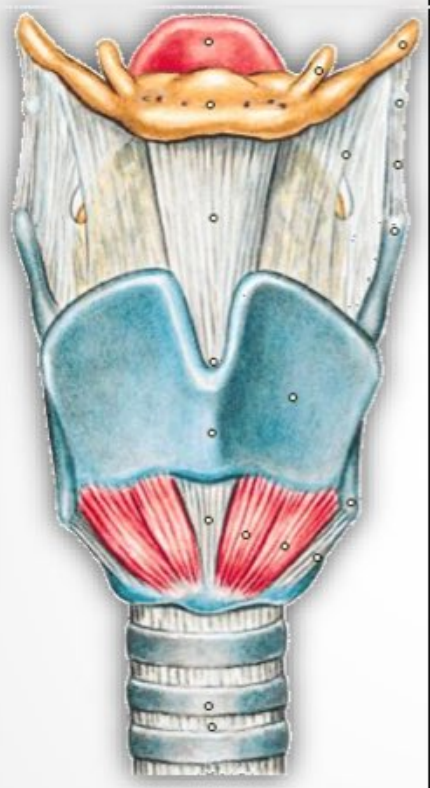
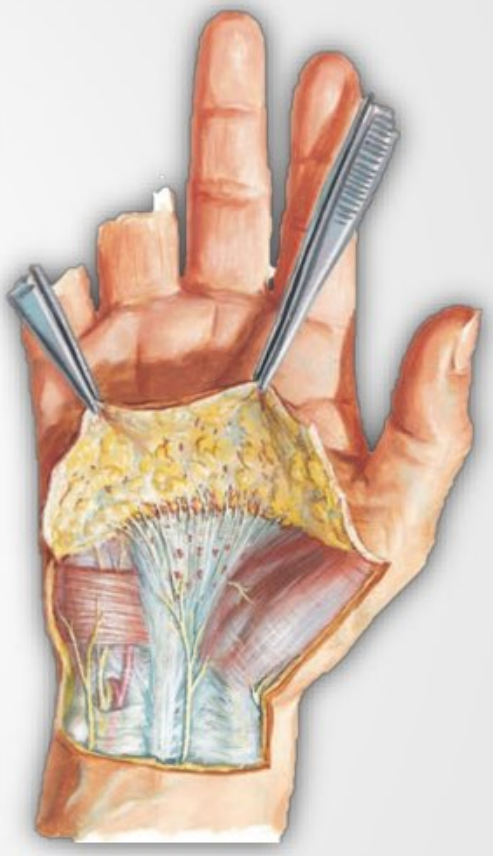
Remarque:
Quelquefois,
l'insertion d'origine
peut devenir
mobile

= distale = mobile

Insertion de terminaison

L'**insertion musculaire** peut se faire sur:

4 sites

Osseuse	Cartilagineuse	Aponévrotique	Cutanée
			
Muscle ilio-psoas	Muscle crico-thyroïdien	Abducteur de l'hallux	Muscle court palmaire

Nombre

Situation

Division

Insertion



Forme

Synthèse: Nomenclature

Annexes

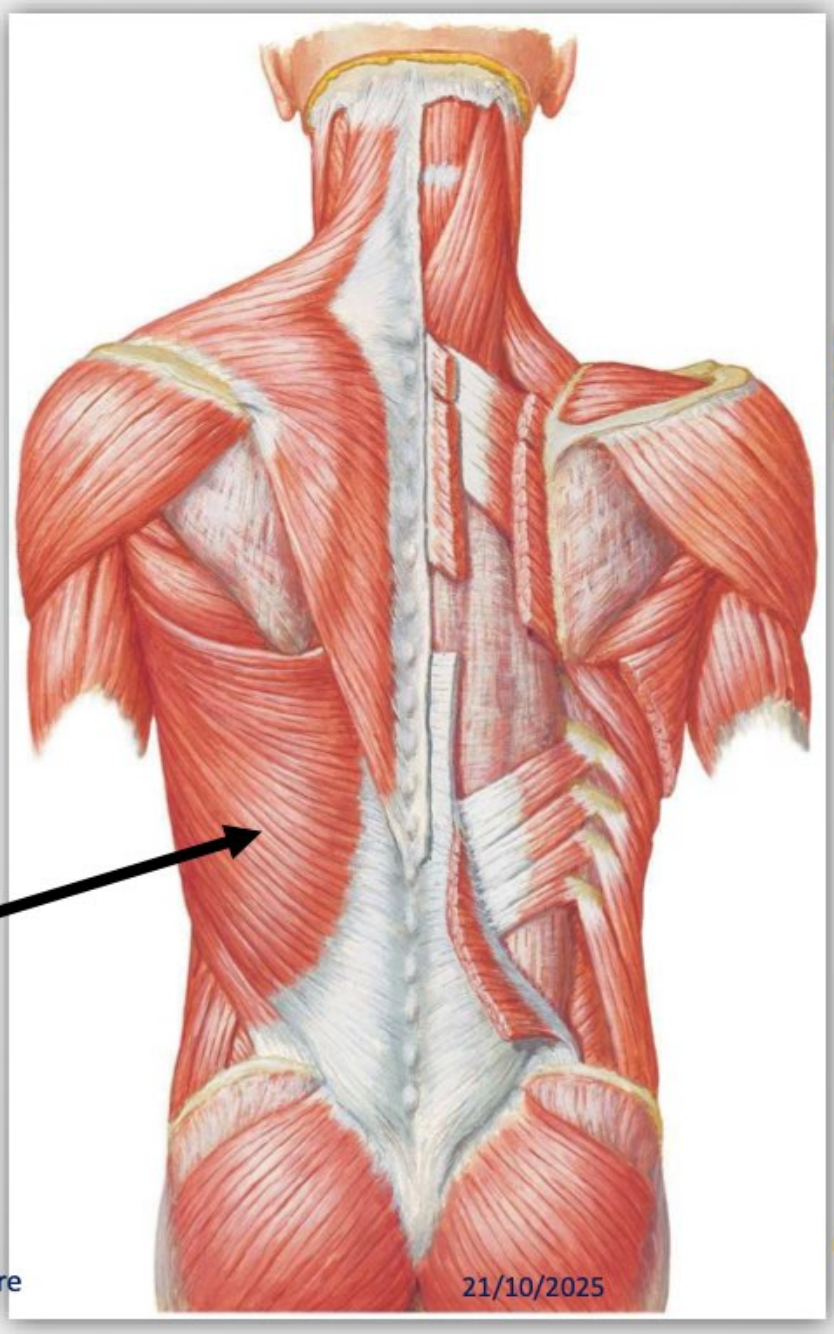
1 Muscle long

- **Longueur** **prédomine** par rapport à **la largeur** et **épaisseur**
- Ex: **sartorius**



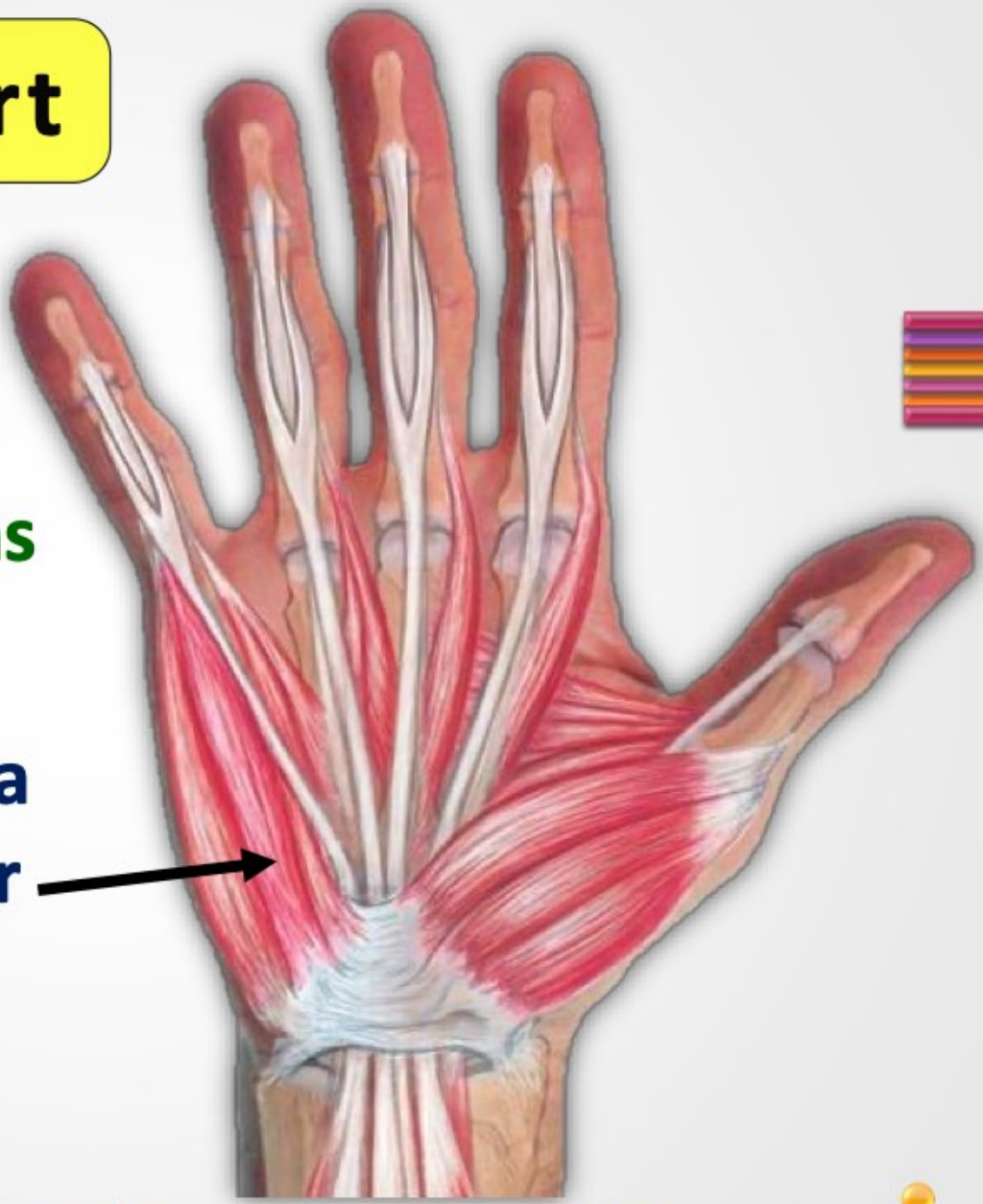
2 Muscle plat

- **Longueur** et **largeur** **prédominent** par rapport à **l'épaisseur**
- Ex: **grand dorsal**



3 Muscle court

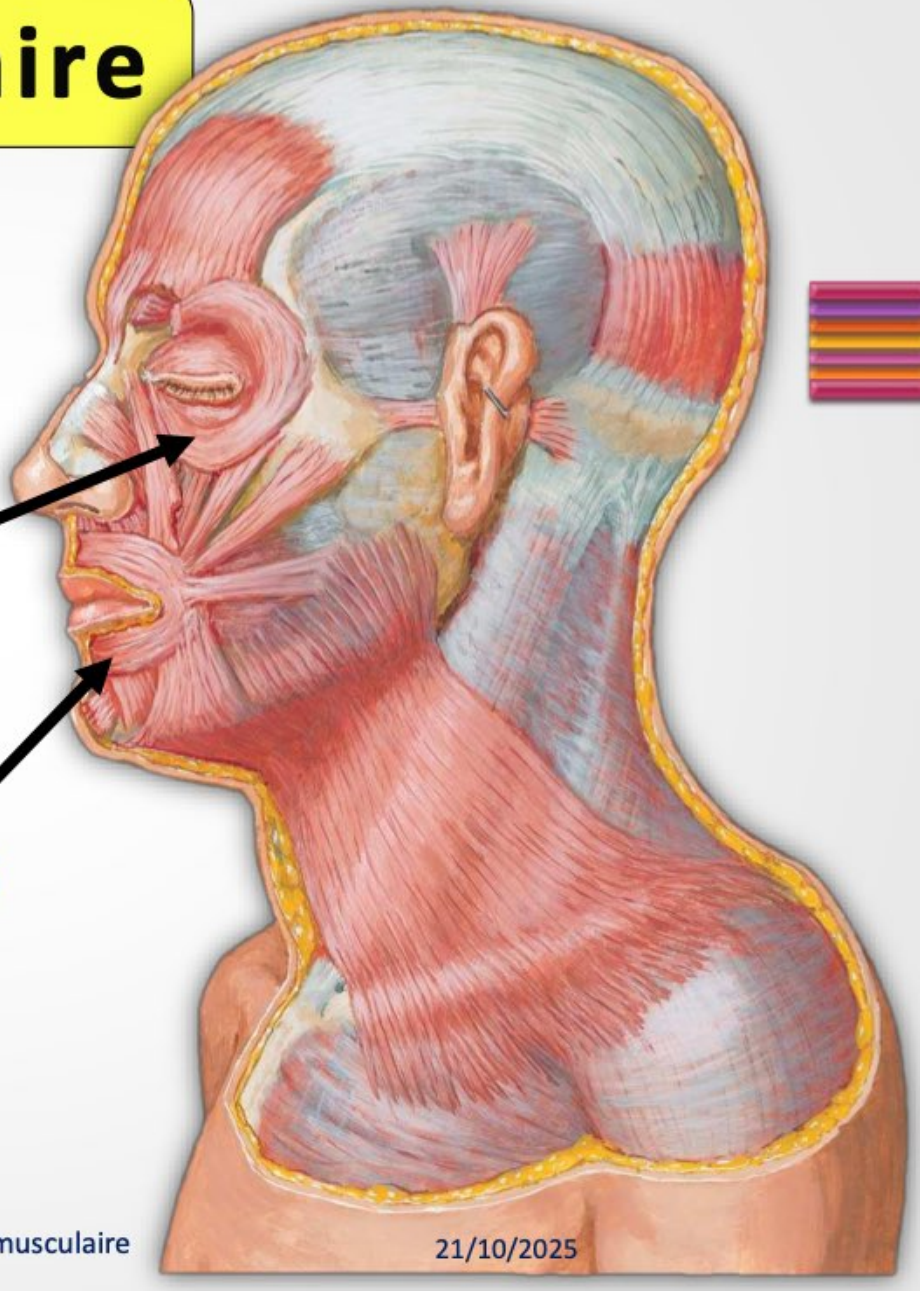
- **Les 3 dimensions** sont **réduites**
- Ex: **muscles de la loge hypothénar**



4 Muscle annulaire

- Situé:
  autour des orifices
- Ex: **orbiculaire des paupières**

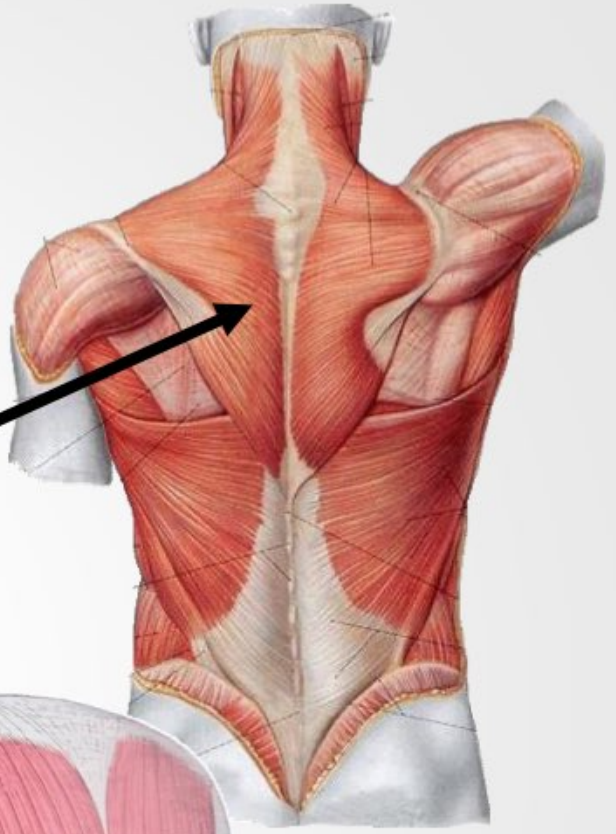
  orbiculaire des lèvres

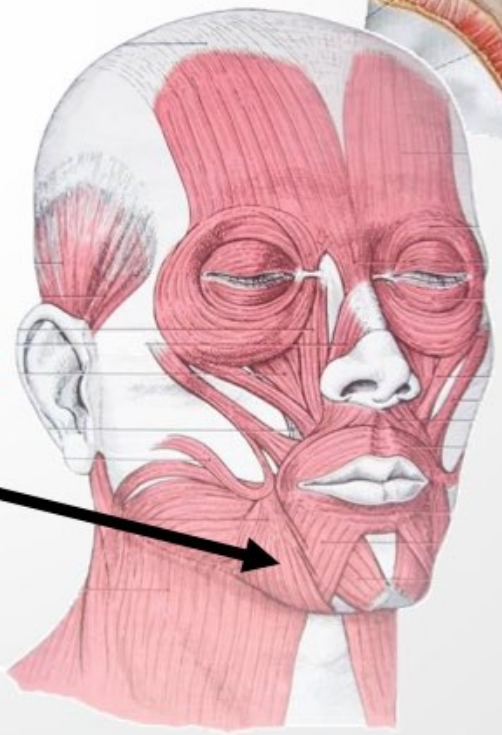


5 Muscle géométrique



- Trapèze
- Carré pronateur
- Triangulaire des lèvres





Nombre

Situation

Division

Insertion

Forme



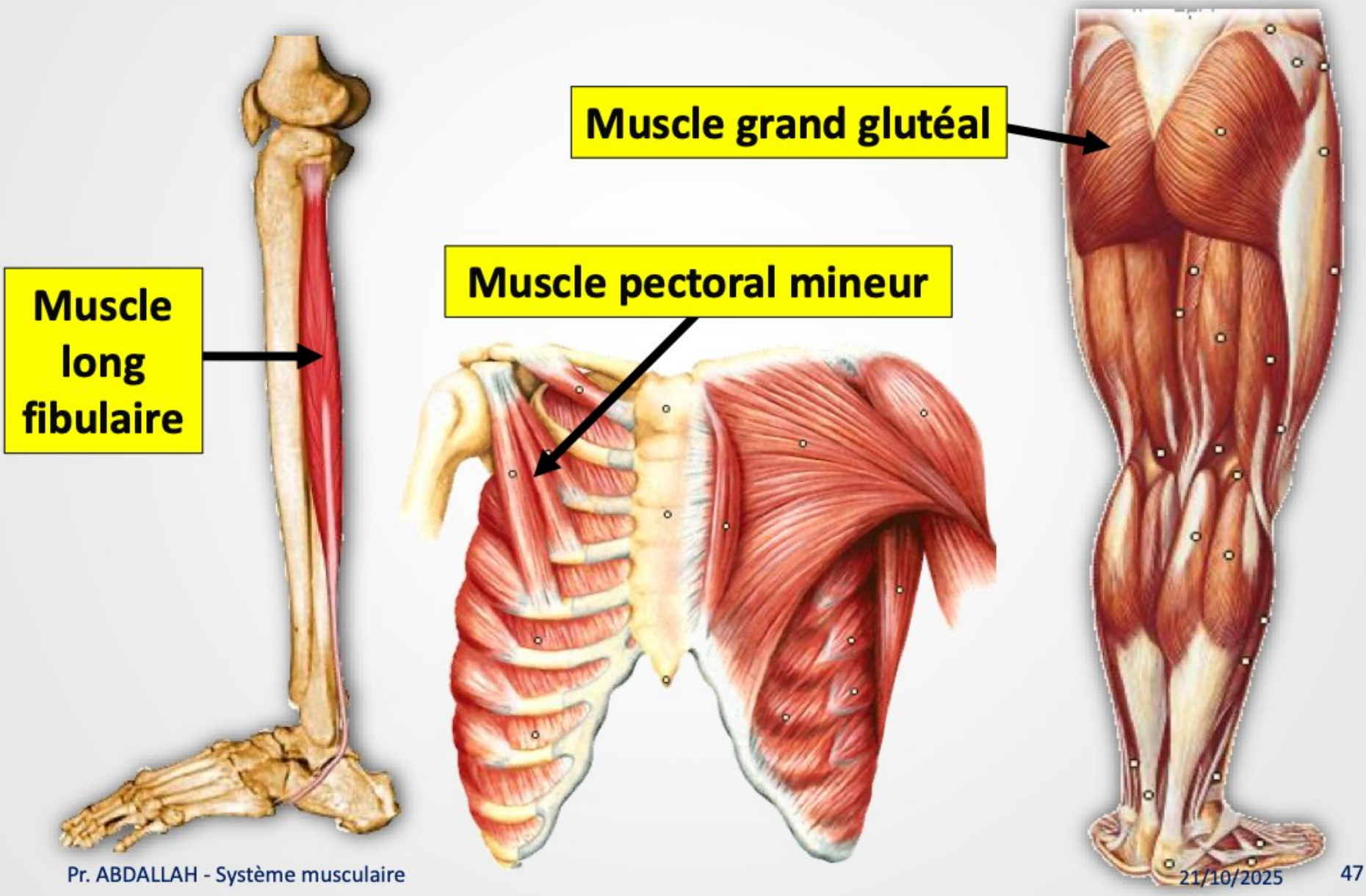
Synthèse: Nomenclature

Annexes

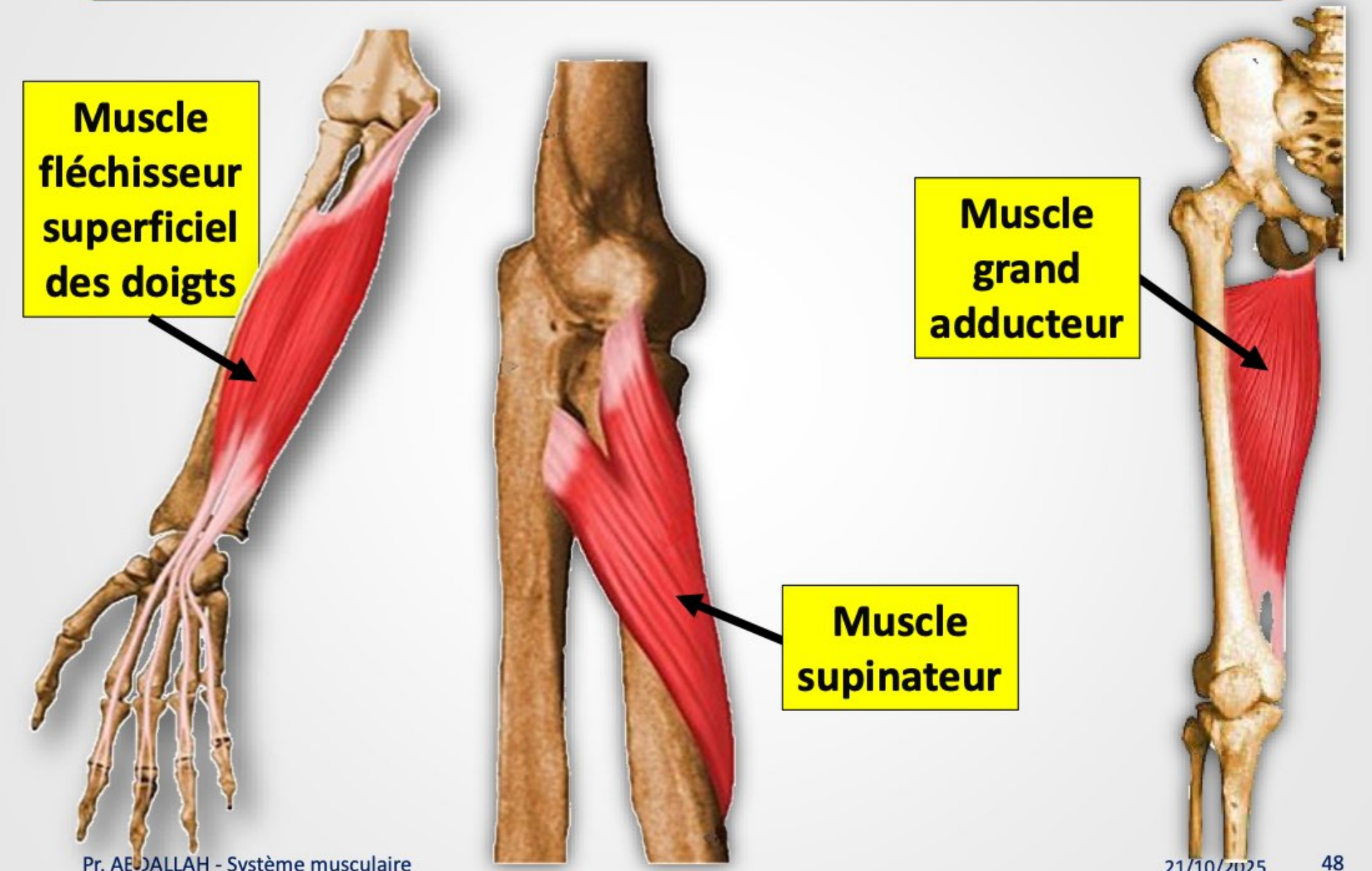
Basée sur:

critères morphologiques et **fonctionnelles**

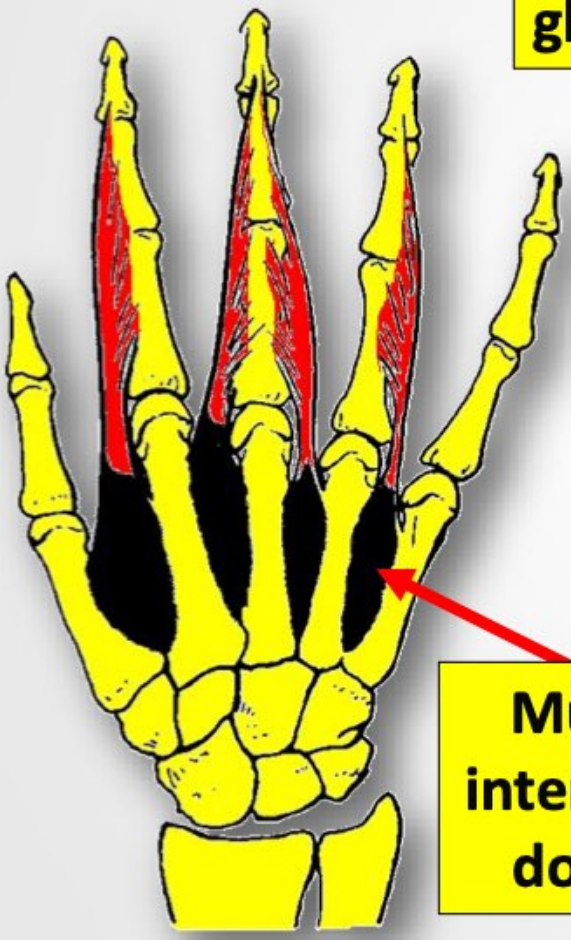
(1) Critère de taille



(2) Critère d'action



(3) Critère de situation



Muscles interosseux dorsaux

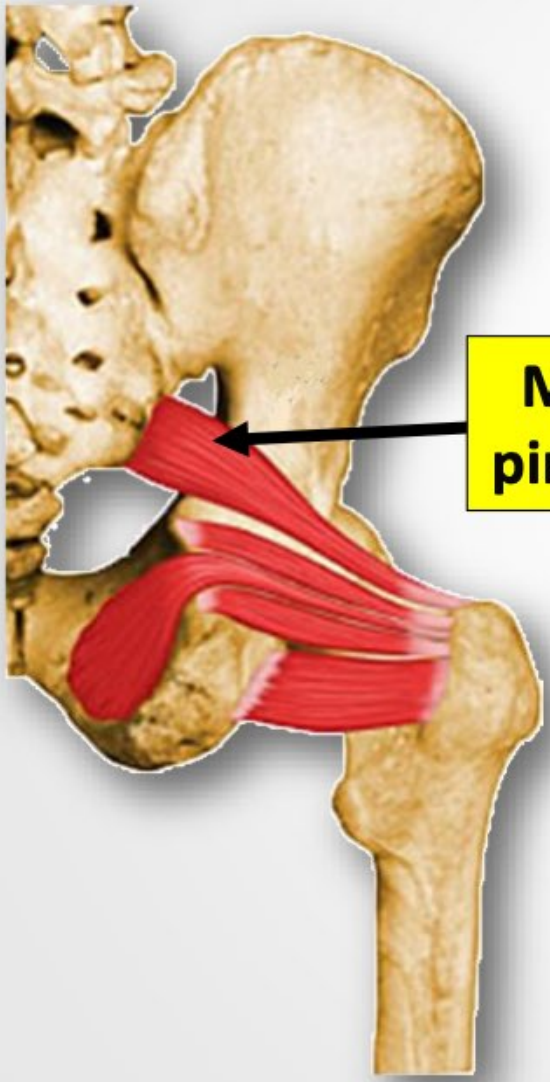


Muscles glutéaux

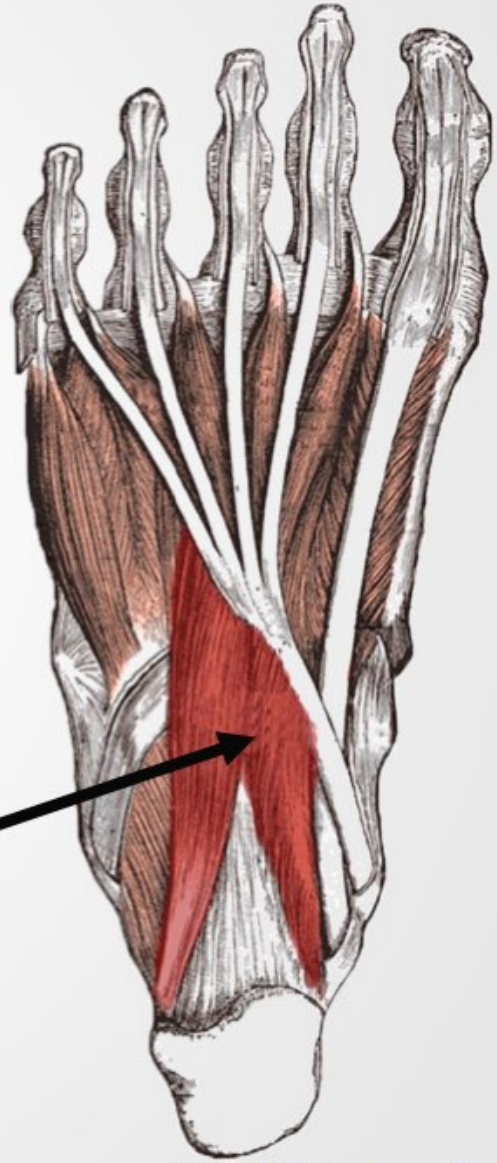


Muscle brachial

(4) Critère de forme

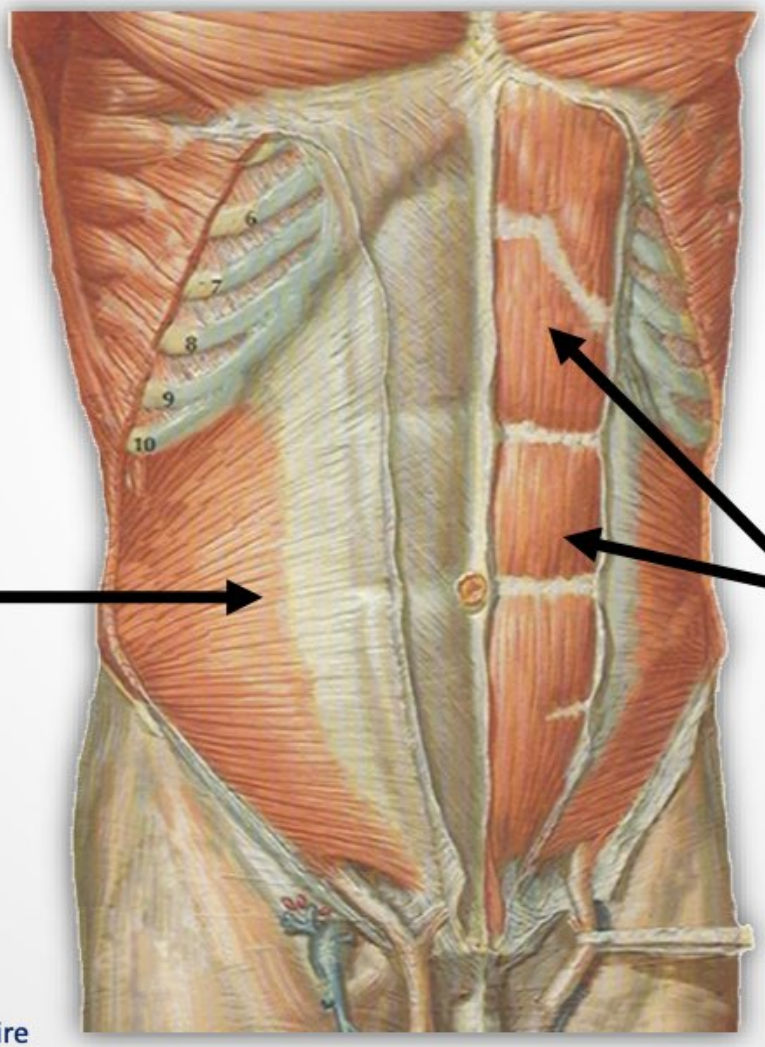


Muscle piriforme



Muscle carré plantaire

(5) Critère de direction des fibres musculaires



Muscle oblique interne

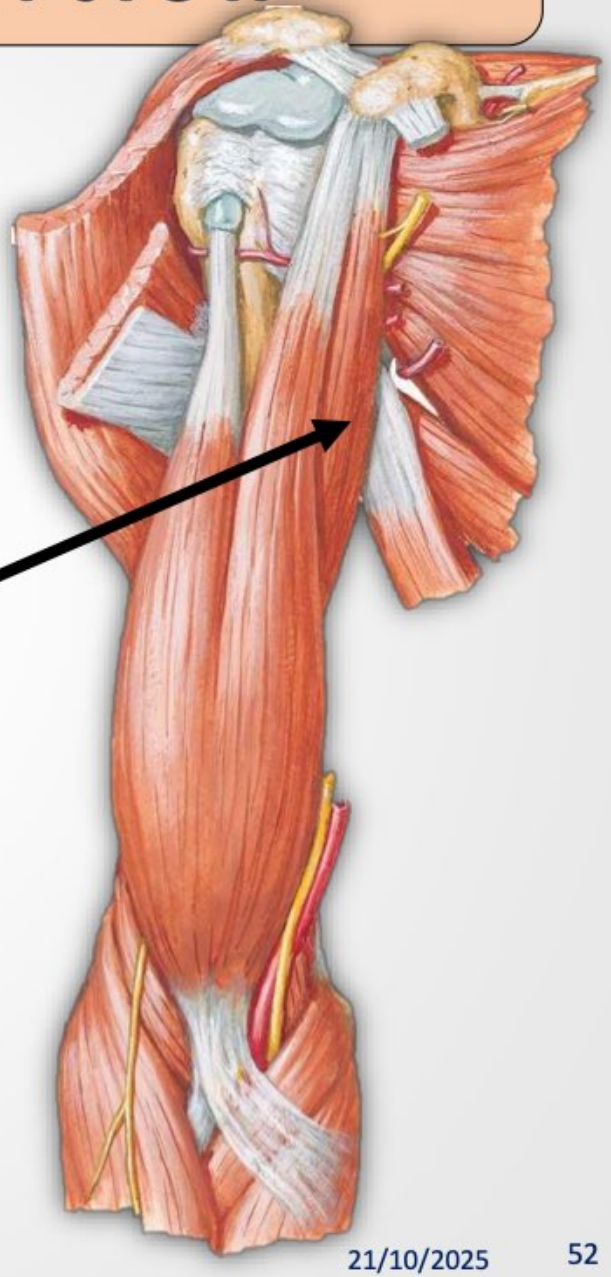
Muscle droit de l'abdomen

(6) Critère d'insertion

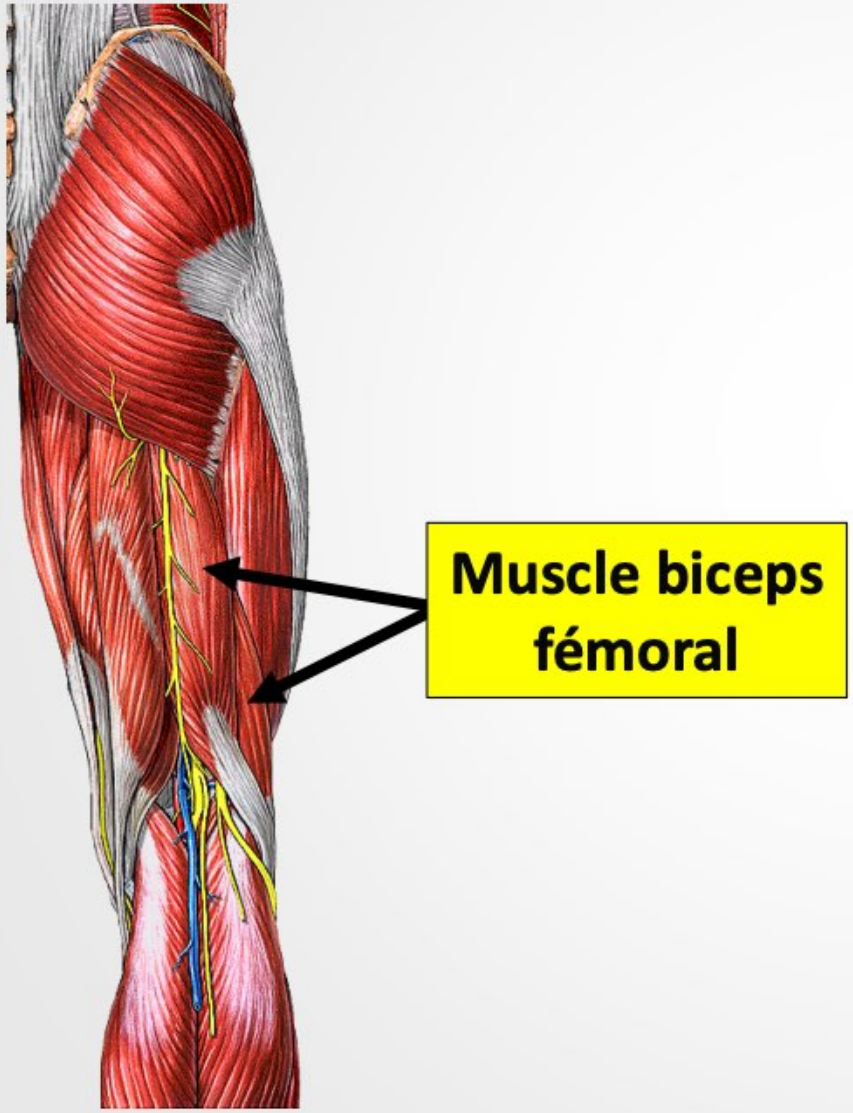


Muscle brachioradial

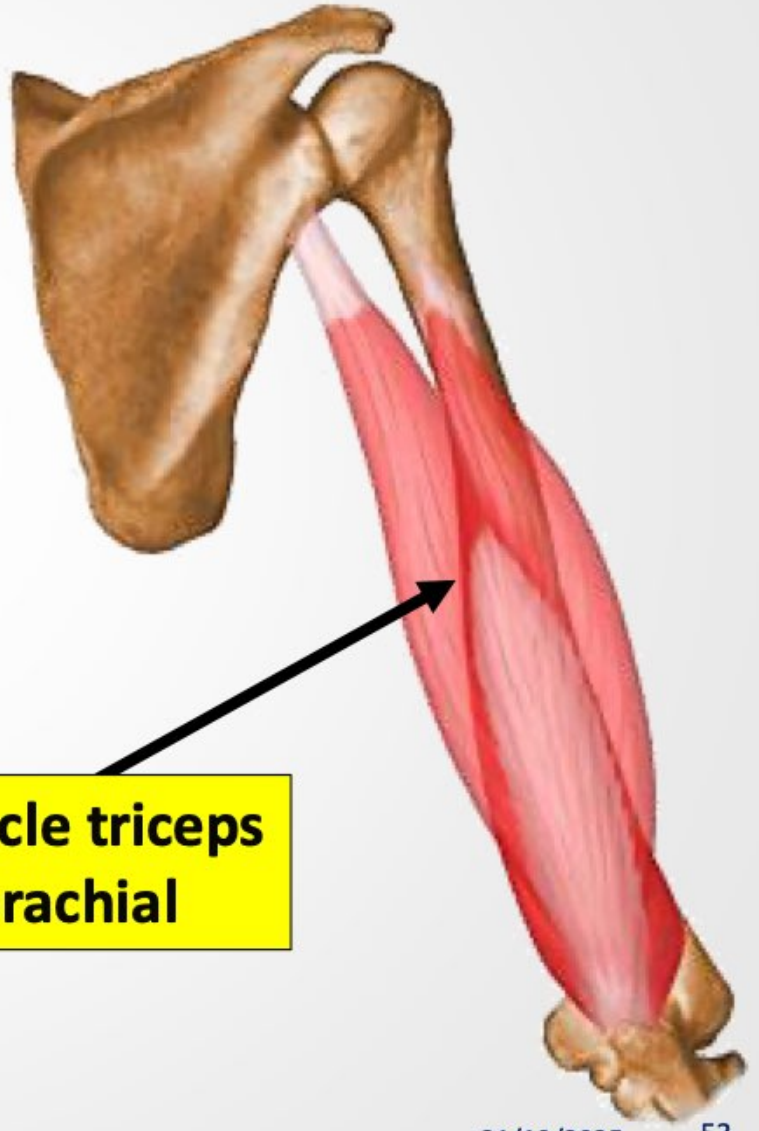
Muscle coracobrachial



(7) Critère de nombre de chefs



Muscle biceps fémoral



Muscle triceps brachial

Nombre

Situation

Division

Insertion

Forme

Synthèse: Nomenclature

Annexes

= **structures anatomiques utiles** au muscle,
facilitant son action

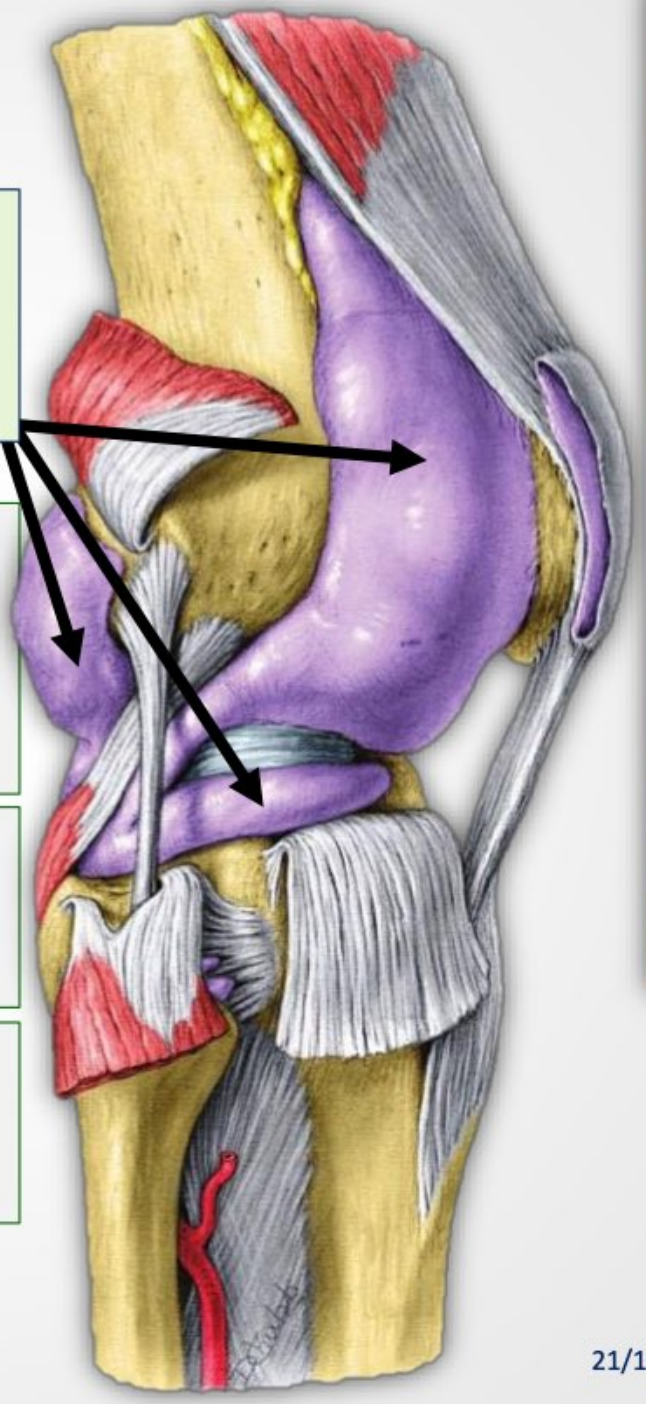
On compte **4**…

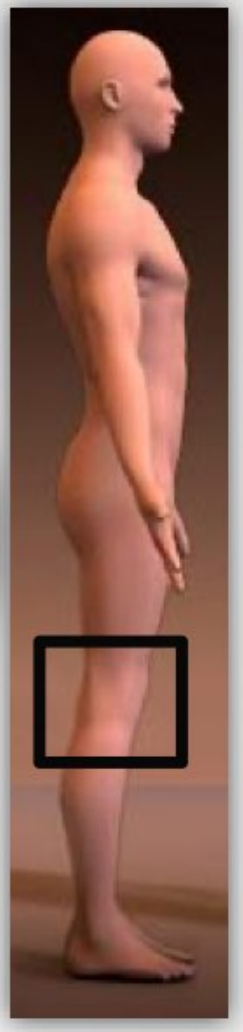
(1) Bourses séreuses (synoviales)

= **poches membraneuses** remplies de **synovie**

Situées au voisinage des articulations

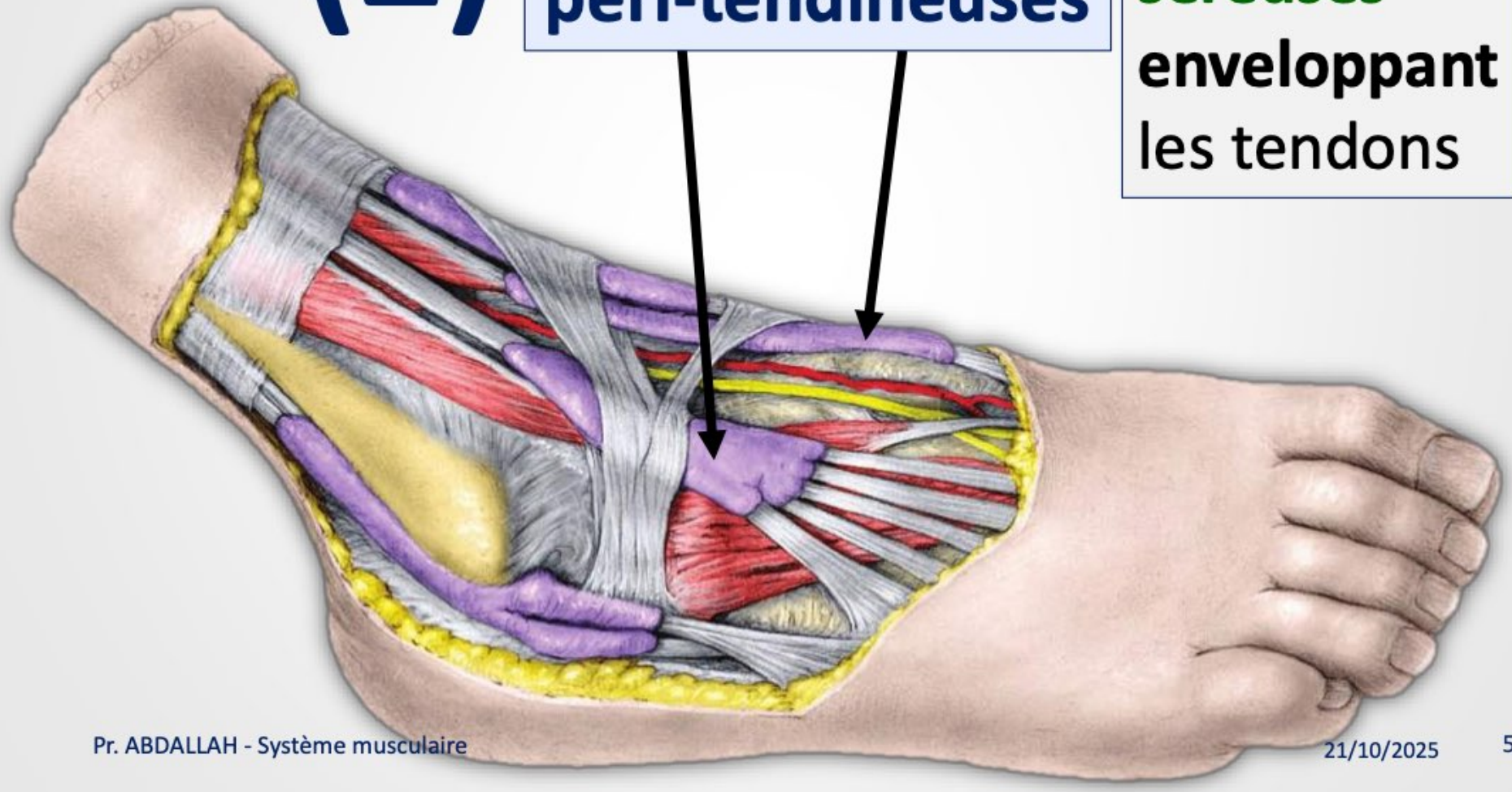
Facilitent: **glissement des muscles**





(2) Gaines séreuses péri-tendineuses

= **bourses séreuses enveloppant** les tendons

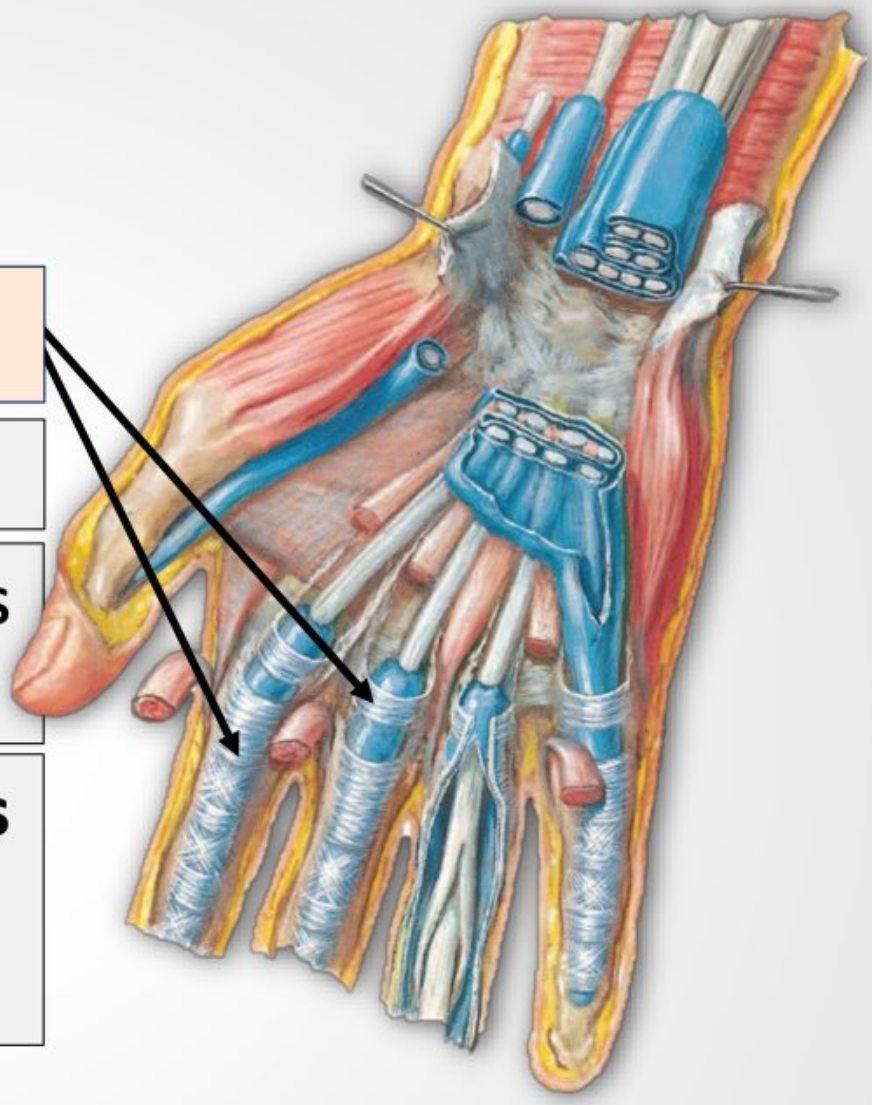


(3) Gaines fibreuses

= **gouttières fibreuses**

Délimitent avec **l'os** des **canaux ostéo-fibreux**

Où **glissent les tendons** enveloppés par des **gaines séreuses**

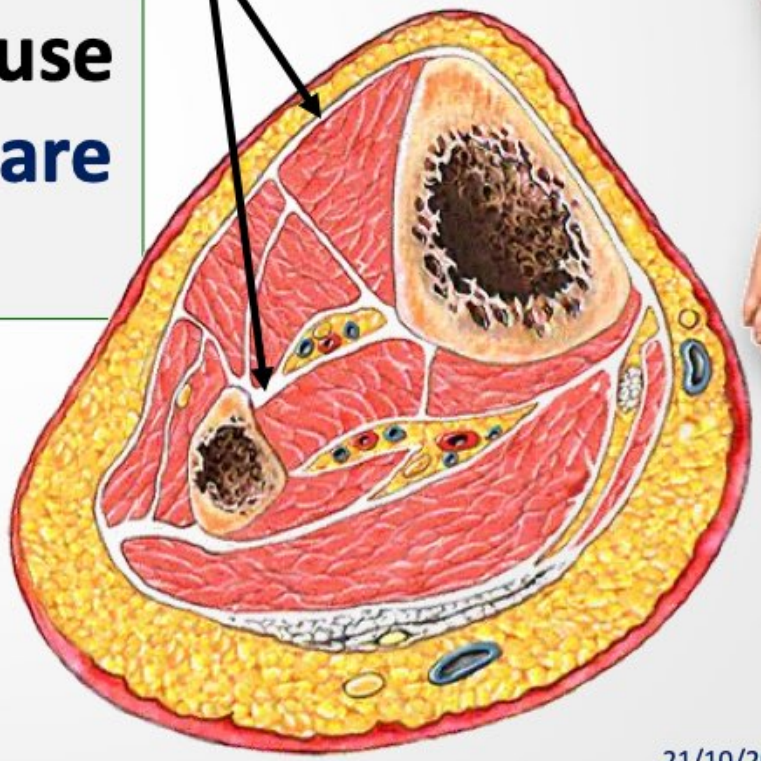


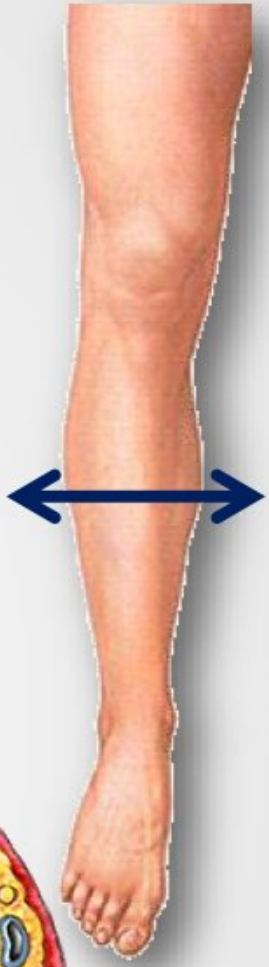


(4) Fascia musculaire

(aponévrose de revêtement)

= **membrane fibreuse** qui **engaine** et **sépare les muscles**



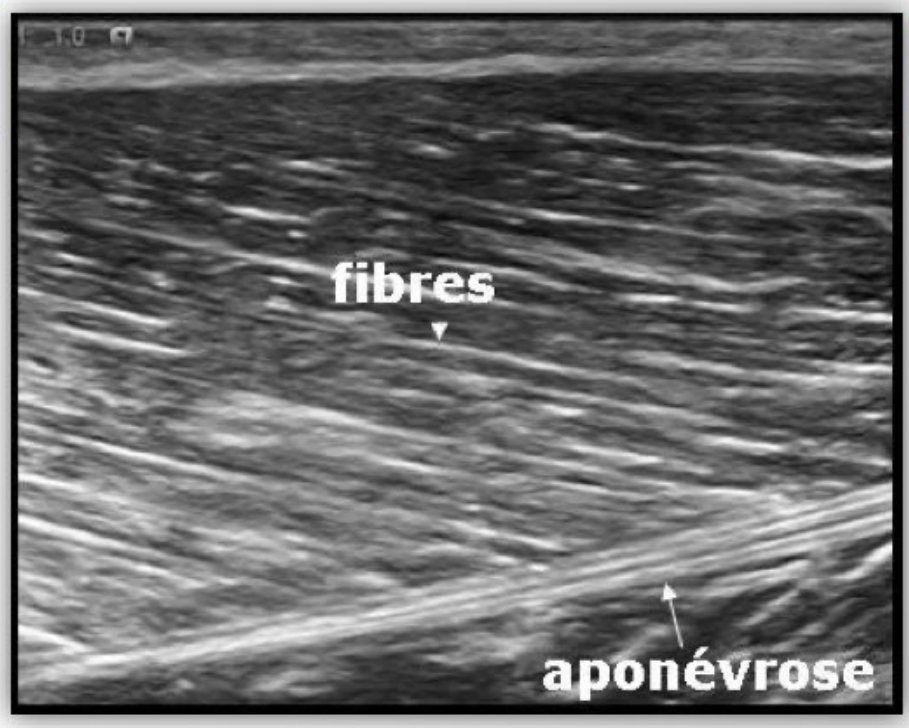


Myologie clinique



Exploration musculaire

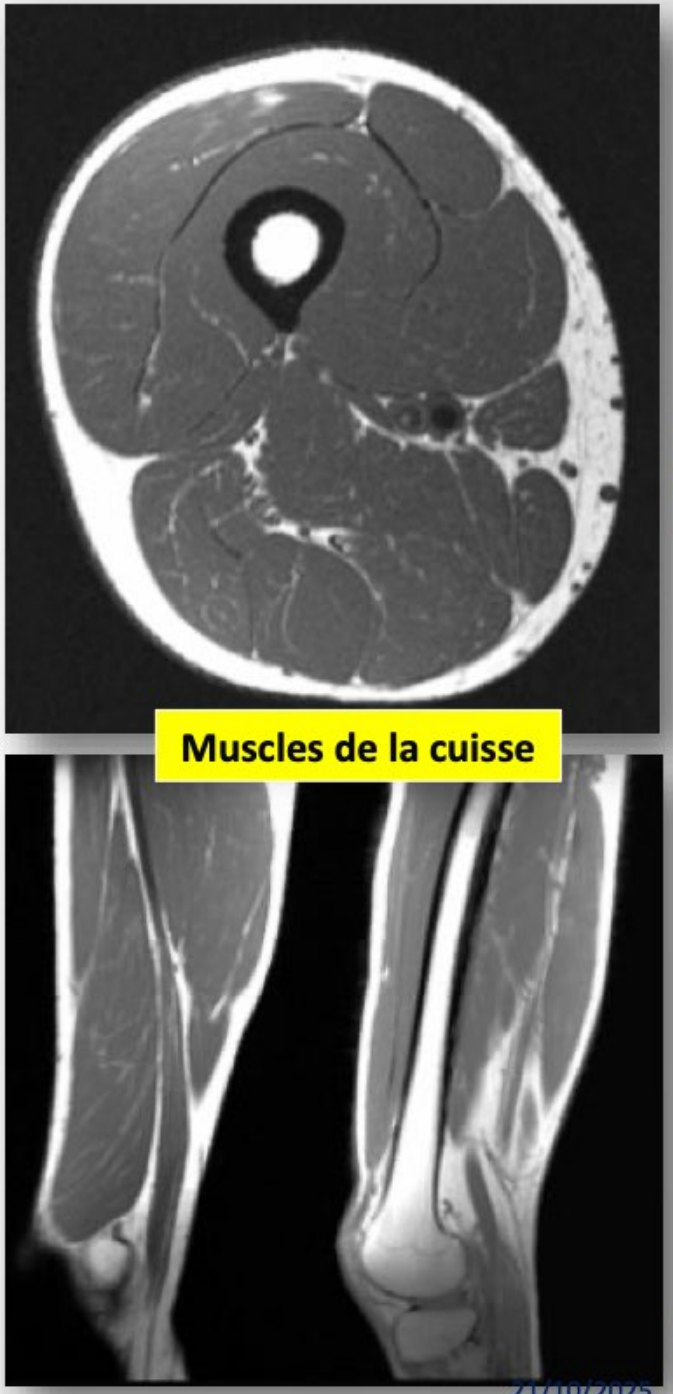
(1) Echographie



(2) Tomodensitométrie (TDM)



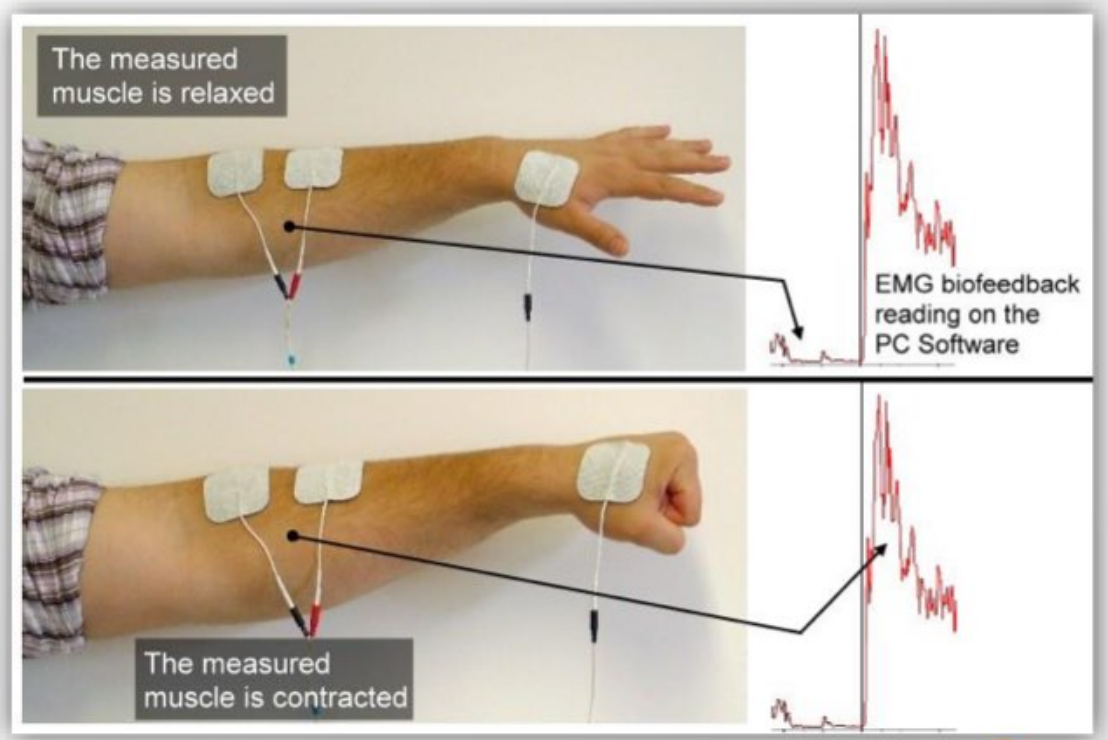
(3) Imagerie par résonance magnétique (IRM)



Muscles de la cuisse

(4) Électromyogramme (EMG)

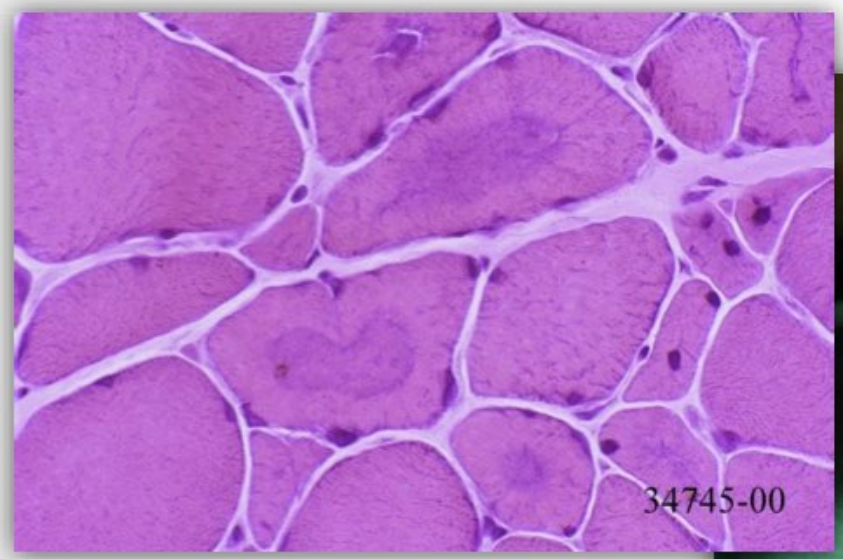
= technique enregistrant **l'activité électrique** du muscle



(5) Biopsie musculaire

= **prélèvement** musculaire







(6) Testing musculaire

= examen évaluant **la force d'un muscle** ou **d'un groupe musculaire**

Coté sur une échelle: de **0** à **5**







Pathologie musculaire

(1) Crampe

= **contraction musculaire brutale, involontaire, douloureuse** et **transitoire**, qui **cesse spontanément**



(2) Courbature

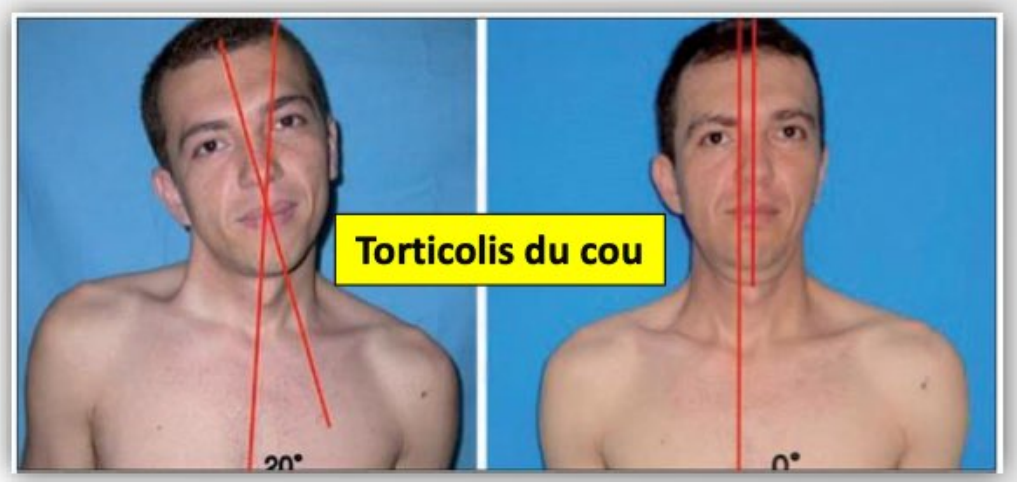
= douleurs musculaires diffuses





(3) Contracture

= contraction musculaire **involontaire**, **douloureuse** et **permanente**

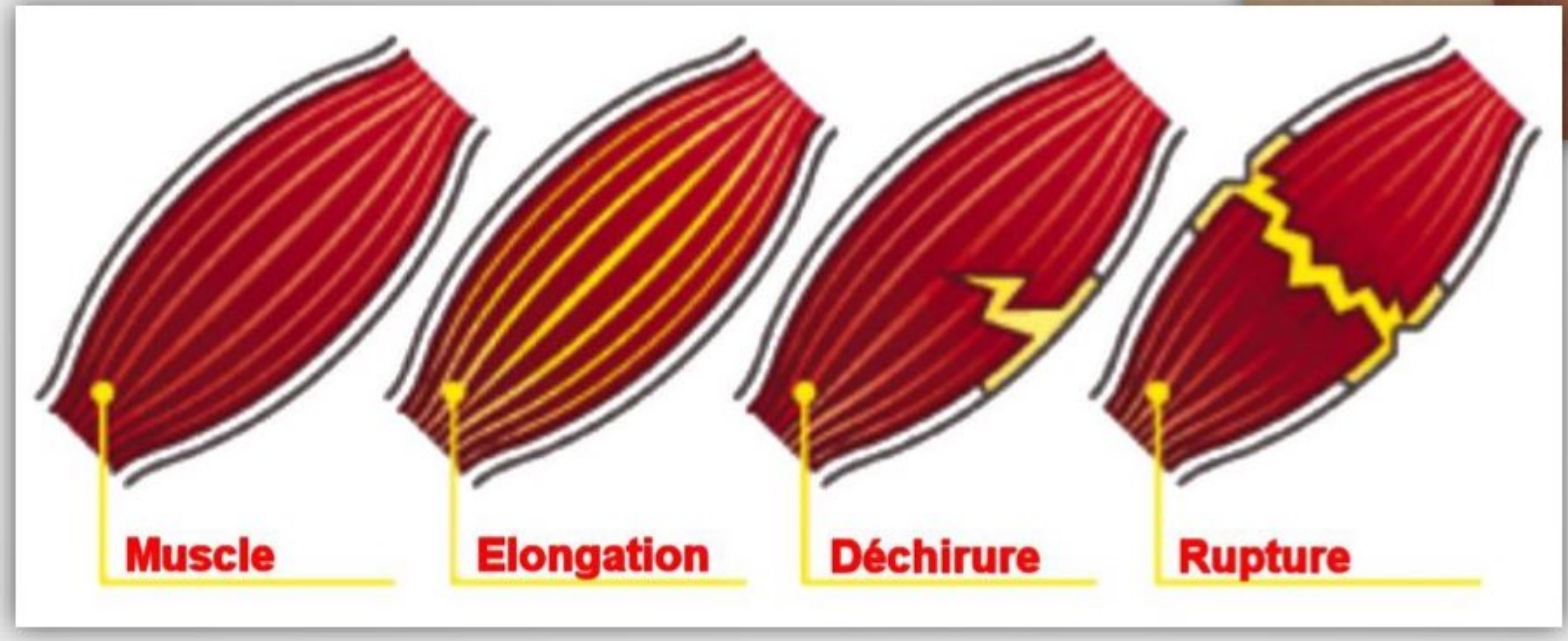




(4) Déchirure (claquage)

= **rupture musculaire** suite à un étirement excessif





(5) Bursite

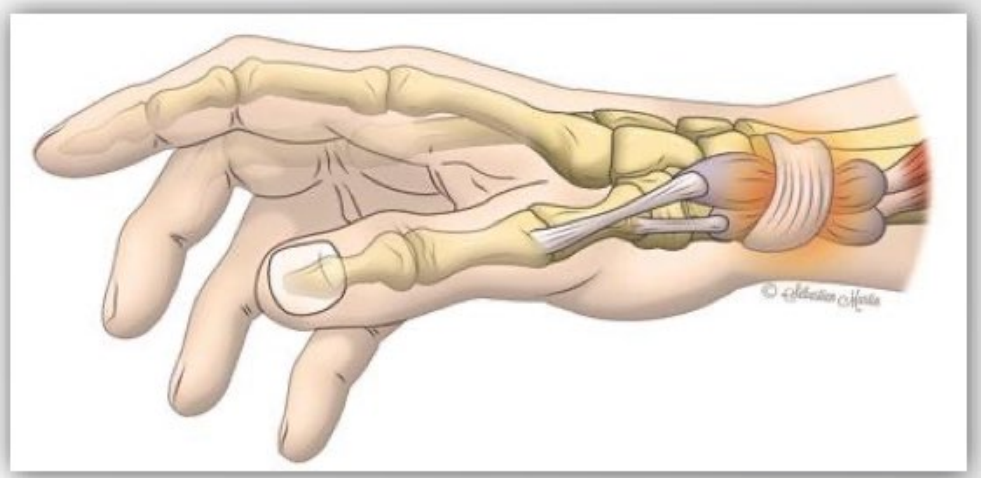
= inflammation des bourses séreuses


Bursite du coude


Bursite rétro-achilléenne

(6) Ténosynovite

= inflammation des gaines synoviales des tendons





(7) Syndrome des loges

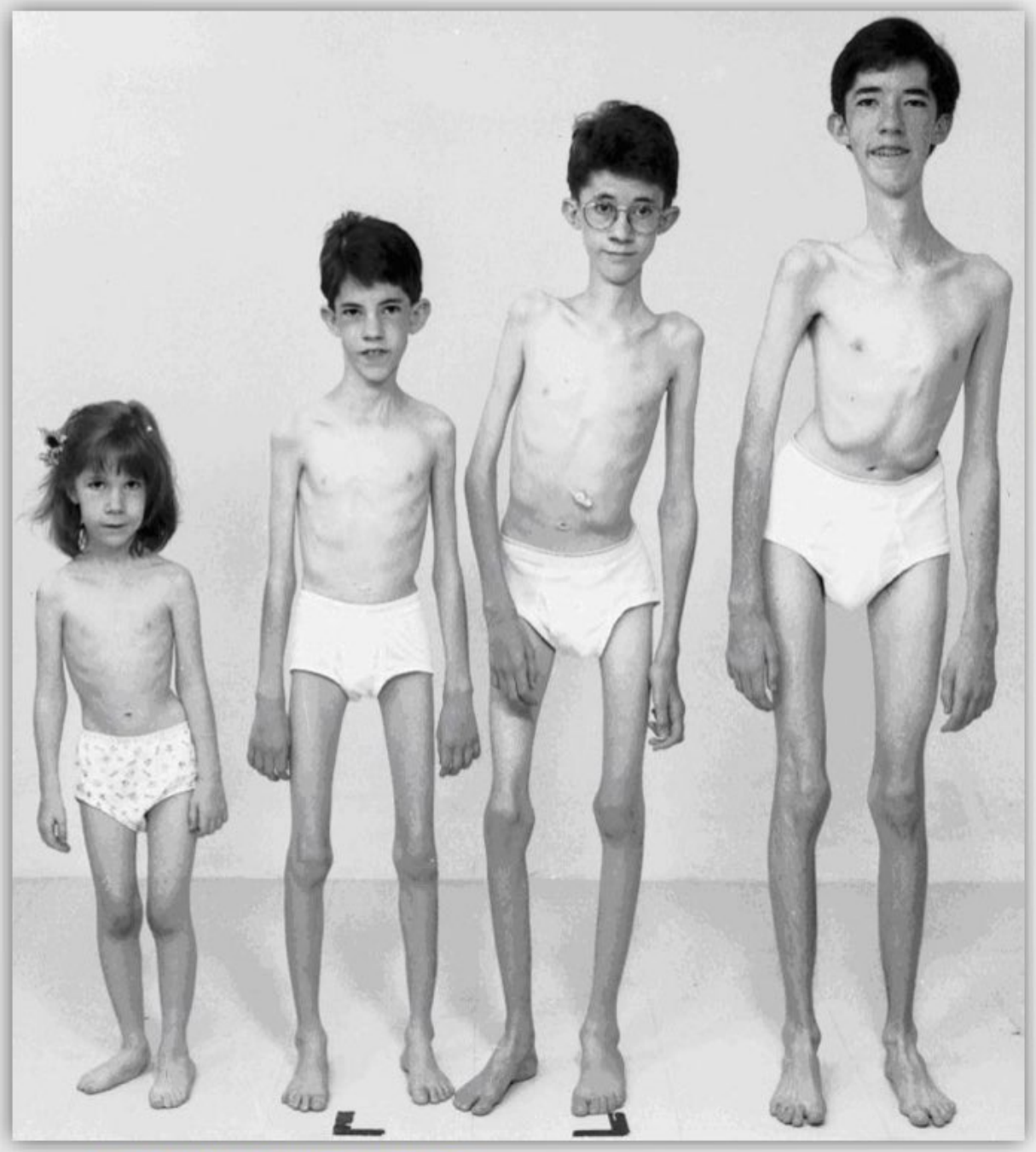
= **ischémie musculaire** par augmentation de pression dans une loge anatomique



(8) Myopathies congénitales


Myopathie facio-scapulo-humérale


Myopathie de Duchenne



Myopathies congénitales



Bonne révision